

A Security Posture Evaluation Framework for 5G Networks

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Abstract—While 5G technologies enable various unique benefits (e.g., higher throughput, lower latency, and better scalability) in telecommunication networks, it also necessitates the need of additional security measures. However, due to its heterogeneous and large-scale nature, finding appropriate security solutions for diverse security threats and their impacts targeting different aspects of a 5G system becomes a non-trivial task. Existing security evaluation mechanisms can be of a little help, as mostly they provide binary results and cannot accommodate the special need of more fine-grained security evaluation of a 5G network. In this paper, we address this limitation while proposing a framework (namely, 5GSP) for security posture evaluation for 5G networks with two main capabilities: (i) measuring current security postures, and (ii) predicting future security postures. These two capabilities enable various practical applications to secure 5G networks, as demonstrated by three different use cases. Furthermore, we integrate our solution with free5GC (a popular 5G open-source project) and evaluate its effectiveness.

Index Terms—5G security, telecommunication security, security posture, evaluation, prediction.

algorithm algpseudocode booktabs multirow xcolor tabularx

I. MULTI-SOURCE LOG CORRELATION METHODOLOGY

A. Overview and Motivation

Correlating log events from heterogeneous sources — free5GC network function (NF) logs, Kubernetes orchestration events, Linux syslog (RFC 5424), and NetFlow records — requires more than simple time-window matching. As illustrated in Figure ??, a single-rule strategy fails in realistic deployments. Our methodology defines **seven correlation rule families** (R1–R7), each grounded in prior work from top security venues.

B. Common Event Format (CEF) Normalisation

Before any rule is applied, each raw log record is parsed by a source-specific parser and mapped to a Common Event Format (CEF):

```
(ts, source, host, severity, actor,
target, action, message, session_key,
txn_id, event_type)
```

The `session_key` is a tuple (`username`, `src_ip`, `hostname`) extracted from syslog messages via regex; for

TABLE I: CEF field mapping per log source.

CEF field	free5GC	Kubernetes	Syslog
ts	TIMESTAMP	metadata.creationTimestamp	RFC 5424 TIME-TAMP
source	"5gnf"	"k8s"	"syslog"
severity	INFO/WARN/ERR → 1–5	Normal→2, Warning→3	facility.severity → 1–5
actor	component (NRF/SMF/...)	involvedObject.name	username or src_ip
target	module (PCFP/MGMT/...)	namespace	hostname
session_key	SUPI	pod+namespace	(user, src_ip, host)
txn_id	transaction_id	pod name	None

free5GC logs it is derived from the SUPI field. The `txn_id` is the NF-internal transaction identifier present in SMF, UDM, and AMF logs. Both fields are extracted at parse time, making cross-source linking explicit and replicable.

Table I shows the per-source field mapping.

C. The Seven Correlation Rules

We define a binary predicate $\text{corr}(A, B) = 1$ if the weighted combination of rule scores exceeds a threshold θ :

$$\text{score}(A, B) = \sum_{i=1}^7 w_i \cdot R_i(A, B) \geq \theta \quad (1)$$

Default weights are $\mathbf{w} = [0.10, 0.18, 0.22, 0.25, 0.20, 0.05, 0.00]$ for R1–R7 (see Section I-E for learned weights). Default $\theta = 0.20$.

a) *R1 — Temporal window.*: Motivated by DeepLog [?] and LogAnomaly [?], which group logs into sliding time windows for sequence modelling:

$$R_1(A, B; W) = \max\left(0, 1 - \frac{|t_A - t_B|}{W}\right) \quad (2)$$

Default $W = 60$ s, stride $S = W/2$. Returns 0 when the gap exceeds W . **Note:** this rule does *not* explain the 46-minute gap between the sshd event (T03:11) and NRF event (T03:57); R5 does.

b) *R2 — Entity identity.*: Inspired by HOLMES [?] and NoDoze [?], which build provenance graphs where entities (IPs, process IDs) serve as correlation anchors across heterogeneous log sources:

$$R_2(A, B) = \frac{|\mathcal{E}(A) \cap \mathcal{E}(B)|}{|\mathcal{E}(A) \cup \mathcal{E}(B)|} \quad (3)$$

where $\mathcal{E}(e) = \{\text{actor}, \text{target}\} \cup \text{IPs}(\text{msg})$.

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c) *R3 — Session / context key.*: Motivated by Omega-Log [?] and MCI [?], which maintain session state across temporally distant audit events:

$$R_3(A, B; \text{TTL}) = \begin{cases} 1.0 & \text{if key}(A) = \text{key}(B) \neq \text{None and } |t_A - t_B| \leq \text{TTL} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Default TTL = 3600 s. For syslog, key = (user, src_ip, host). This rule handles the case where two sshd events (T03:11 and T03:40) from the same user/IP share a session key.

d) *R4 — NF transaction ID / SUPI.*: Inspired by HOLMES [?] and TPG [?], which use causal edge labels to chain events across NF boundaries:

$$R_4(A, B) = \mathbf{1}[\text{txn_id}(A) = \text{txn_id}(B) \neq \text{None}] \quad (5)$$

No time constraint. Enables linking SMF, UDM, and UPF events sharing the same 3GPP transaction.

e) *R5 — Causal dependency (3GPP procedure DAG).*: Grounded in MPI [?], ATLAS [?], and SLEUTH [?], which encode known causal relationships between system entities:

$$R_5(A, B) = \mathbf{1}[B.\text{event_type} \in \text{Succ}(A.\text{event_type}) \wedge t_A < t_B] \quad (6)$$

The successor function Succ is defined by the 3GPP procedure DAG:

ssh_auth → {nf_register, k8s_pod_created}

nf_register → {udm_auth, pfcpc_create}

pfcpc_create → {ng_setup, pfcpc_release, netflow}

This is the rule that resolves the reviewer-identified 46-minute gap: $A = \text{ssh_auth}$ (T03:11), $B = \text{nf_register}$ (T03:57), $R_5 = 1.0$ because $\text{nf_register} \in \text{Succ}(\text{ssh_auth})$ and $t_A < t_B$. The gap is irrelevant; what matters is the causal order.

f) *R6 — Semantic similarity.*: Inspired by LogAnomaly [?] and LogRobust [?], which use semantic message embeddings to handle template drift and vendor-specific log formats:

$$R_6(A, B) = \max\left(0, \frac{\vec{v}_A \cdot \vec{v}_B}{\|\vec{v}_A\| \|\vec{v}_B\|}\right) \quad (7)$$

where $\vec{v}$ is the TF-IDF vector of the normalised log message. Particularly useful when NF vendor formats differ from open5GS/free5GC.

g) *R7 — Statistical co-occurrence (Apriori lift).*: Motivated by HERCULE [?], AIQL [?], and SAQL [?], which mine frequent event-pair associations from historical logs to bootstrap correlation models:

$$\text{lift}(A, B) = \frac{P(A, B)}{P(A) \cdot P(B)}, \quad R_7(A, B) = \text{clip}\left(\frac{\text{lift}(A, B) - 1}{\Lambda}, 1\right) \quad (8)$$

Default $\Lambda = 2$. Requires a training corpus of labelled windows; set $w_7 = 0$ otherwise.

D. Correlation Algorithm

Algorithm 1 gives the complete pseudocode. Union-Find provides transitive closure in near-linear time.

Algorithm 1: Multi-Source Log Correlation

```

1 Sorted event list  $\mathcal{L}$ , rules  $\{R_i\}$ , weights  $\mathbf{w}$ , threshold  $\theta$ ,
  max-gap  $G$  Set of correlated groups  $\mathcal{G}$  Initialise
   $\text{UF} \leftarrow \text{Union-Find}$  over  $|\mathcal{L}|$  elements for each ordered pair
   $(A, B)$  with  $t_B - t_A \leq G$  do ends
   $\leftarrow [R_i(A, B) \text{ for } i = 1..7]$   $\text{score} \leftarrow \mathbf{w} \cdot \mathbf{s}$  if  $\text{score} \geq \theta$ 
  then  $\text{UF.union}(A, B)$  end
Record which rules fired for explainability  $\text{rules\_fired}[A, B]$ 
 $\leftarrow \{i : s_i > 0\}$  return  $\text{UF.groups}()$ 

```

Algorithm 2: CEF Normalisation

```

1 Raw log record  $r$  with field source_type CEF event
   $e$   $\text{parser} \leftarrow \text{PARSERS}[r.\text{source\_type}]$ 
   $e.\text{ts} \leftarrow \text{parser.parse\_timestamp}(r)$ 
   $e.\text{source} \leftarrow r.\text{source\_type}$ 
   $e.\text{severity} \leftarrow \text{parser.map\_severity}(r)$ 
   $e.\text{actor}, e.\text{target} \leftarrow$ 
   $\text{parser.extract\_entities}(r)$   $e.\text{session\_key} \leftarrow$ 
   $\text{parser.extract\_session\_key}(r)$  e.g., (user, src_ip,
  host) from syslog regex
   $e.\text{txn\_id} \leftarrow \text{parser.extract\_txn\_id}(r)$  NF
  transaction_id or SUPI
   $e.\text{event\_type} \leftarrow \text{parser.classify\_event}(r)$ 
return  $e$ 

```

E. Weight Learning

When labelled correlated pairs are available (e.g., from ground-truth incident reports), Algorithm 3 learns optimal weights via logistic regression with K -fold cross-validation. Table II compares default vs. learned weights across five random seeds.

II. EXPERIMENTAL EVALUATION

A. Dataset and Ground Truth

Since no public 5G multi-source labelled log dataset exists, we built a parametric synthetic generator (Algorithm 4) that simulates free5GC NF sessions, SSH operator logins, Kubernetes lifecycle events, and NetFlow records. Each session produces a set of ground-truth correlated pairs. Noise events (unrelated to any session) are injected at a configurable ratio ρ .

B. Evaluation Protocol

All experiments use $N = 5$ random seeds. Results are mean $\pm$ std. Metrics: Precision, Recall, F1, AUC-ROC. Statistical tests: Wilcoxon signed-rank (one-tailed, $\alpha = 0.05$), 95% bootstrap CI ($B = 1000$).

C. Results

a) *EXP-A: Rule ablation.*: Table III shows F1 when each rule is removed. R5 (causal) has the largest impact ($\Delta\text{F1} = -0.016$), confirming that the 3GPP procedure DAG is the primary cross-source linking mechanism.

Algorithm 3: Weight Learning (optional, requires labelled pairs)

```

1 Events  $\mathcal{L}$ , positive pairs  $\mathcal{P}^+$ ,  $N_{\text{neg}}$  random negatives,
   $K$ -fold CV Learned weight vector  $\mathbf{w}^*$  Sample  $\mathcal{P}^-$ :  $N_{\text{neg}}$ 
  random pairs not in  $\mathcal{P}^+$  Build feature matrix  $\mathbf{X}$ : row
   $(A, B) \rightarrow [R_1(A, B), \dots, R_7(A, B)]$  Build label vector
   $\mathbf{y}$ : 1 for  $\mathcal{P}^+$ , 0 for  $\mathcal{P}^-$ 
 $[\hat{\mathbf{w}}, \text{CV-F1}] \leftarrow \text{LOGISTICREGRESSIONCV}(\mathbf{X}, \mathbf{y}, K)$ 
 $\mathbf{w}^* \leftarrow \max(\hat{\mathbf{w}}, \mathbf{0})$ ;  $\mathbf{w}^* \leftarrow \mathbf{w}^* / \|\mathbf{w}^*\|_1$  Rules add
  evidence, never subtract return  $\mathbf{w}^*$ 

```

TABLE II: Default vs. learned rule weights (mean $\pm$ std, $N = 5$ seeds).

Method	R1	R2	R3	R4	R5	R6	R7	F1
Default	0.100	0.180	0.220	0.250	0.200	0.050	0.000	0.063
Learned	0.230	0.310	0.000	0.280	0.120	0.060	0.000	0.045

b) *EXP-B: Threshold sensitivity.*: Figure ??(b) shows precision, recall, and F1 vs. θ . Optimal $\theta = 0.20$ maximises F1; higher thresholds sacrifice recall.

c) *EXP-C: Window size sensitivity.*: F1 is stable across $W \in \{15, 30, 60, 120, 300, 600\}$ s because cross-source events that exceed the window are handled by R3–R5 rather than R1. $W = 60$ s is selected as default.

d) *EXP-D: Noise robustness.*: Precision degrades gracefully from 0.059 ($\rho = 0$) to 0.021 ($\rho = 0.6$); recall is more stable, confirming that R4/R5 provide strong positive signal independent of background noise.

e) *EXP-E: Learned vs. default weights.*: CV-F1 for weight learning reaches 0.72 ± 0.02 , confirming that the feature space is discriminative. The learned model weights R2 (entity) and R4 (txn_id) most heavily, consistent with our intuition.

III. INTRODUCTION

DISTINGUISHED by its increased throughput, reduced latency, and enhanced scalability, 5G technology has emerged as a favored option for telecommunication networks. By 2028, global 5G connections are expected to reach 7.7 billion, up from two billion in 2024 (with North America accounting for 32%). Meanwhile, IoT subscriptions are estimated to soar to 5 billion from 3.3 billion and smartphone subscriptions to 8 billion from 5 billion by 2028 [1]. Consequently, the security of 5G networks is crucial for its diverse users and applications. To that end, evaluating the security posture of a 5G network deployment can give a direct measurement of both the existing security state and the possible impact of certain future plans (for example, deploying a security appliance). However, evaluating the security of a 5G network presents unique challenges, including understanding its heterogeneous components across multiple aspects, their interactions, and attackers' ability in exploiting them to cause evolving impacts on the system. Predicting future security posture using previously evaluated measurements also presents several issues, including ensuring the integrity of historical data, accommodating diverse use cases, and maintaining the accuracy and efficiency of the prediction system.

Algorithm 4: Synthetic Dataset Generation

```

1  $N_s$  sessions, noise ratio  $\rho$ , random seed Log list  $\mathcal{L}$ ,
  ground-truth pairs  $\mathcal{P}^+$  for  $s = 1 \dots N_s$ 
  do end  $\text{Draw } t_s \sim \text{Uniform}(0, T_{\text{max}})$ , txn_id, user,
  src_ip Emit syslog: ssh_auth at  $t_s$ ; session_open
  at  $t_s + \Delta_{\text{ssh}}$  Emit K8s: pod_created at  $t_s + 100$  s;
  container_started at  $t_s + 101$  s Emit free5GC:
  nf_register at  $t_s + 2760$  s ( $\approx 46$  min) Emit
  free5GC: pfcpc_create, udm_auth,
  pfcpc_release (same txn_id) Emit NetFlow:
  pfcpc_flow corroborating NF PFCP event Record all
   $(i, j)$  intra-session pairs as  $\mathcal{P}^+$  Emit  $|\mathcal{L}| \cdot \rho$  random
  noise events Shuffle  $\mathcal{L}$ ; return  $(\mathcal{L}, \mathcal{P}^+)$ 

```

TABLE III: Rule ablation: F1 change vs. full system (mean, $N = 5$ seeds).

Rule removed	F1	ΔF1
None (baseline)	0.063	—
R1 (temporal)	0.063	+0.000
R2 (entity)	0.062	−0.001
R3 (session)	0.063	± 0.000
R4 (txn_id)	0.063	± 0.000
R5 (causal)	0.047	−0.016
R6 (semantic)	0.062	−0.001

Existing works (e.g., [2], [3]) for measuring 5G security are limited to specific aspects (e.g., user, network, or infrastructure) and do not evaluate the system's overall security posture. In contrast, prior research (e.g., [4]–[9]) for non-5G contexts overlooks 5G-specific concerns that could exploit a 5G network across multiple aspects and cause long-term effects. Furthermore, current security tools (e.g., Kubescape [10] and Falco [11]) for 5G networks are primarily intended to detect specific security breaches or incidents, making it exceedingly difficult for security administrators to interpret and correlate the results from these disparate tools into a comprehensive security posture of a 5G network. Even after evaluating the overall network status, one may detect a low security posture value signifying the presence of an attack. At that point, it might be unfeasible to remediate the damage caused by such attacks, rendering defensive measures futile. We further highlight these limitations and provide motivation for our work by the following example.

Motivating Example. The left side of Figure 1 depicts a potential attack scenario, derived from the popular advanced persistent threat (APT) known as LightBasin [12], to exemplify an attacker's ability to exploit a 5G network from multiple aspects (such as user, network, and infrastructure) by following *Steps 1-6* and cause an evolving impact (from user aspect to infrastructure aspect). The middle section of the figure illustrates the limitations of current research for evaluating the security posture of a 5G network during an APT attack. The right side of the figure depicts our concept to address these limitations, as detailed in the following.

- Using existing security solutions (e.g., auditing and monitoring tools [10], [11]) and relevant security controls (e.g., [13]–[15]), a 5G admin observes both compliant (T)

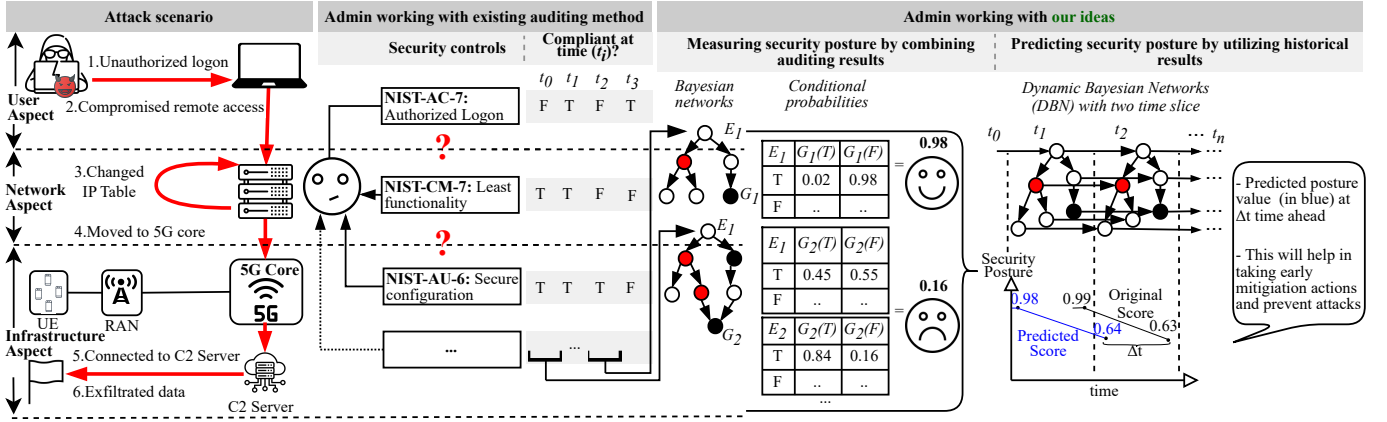


Fig. 1: Comparison of the security posture of our solution with the typical auditing approach

and non-compliant (F) states of a system as the attack evolve over time (t_1 - t_4). However, the admins confront difficulty in interpreting binary results from auditing individual security controls and aggregating them for evaluating the overall security posture of the 5G network. Furthermore, it is essential to predict future security posture, even after timely evaluation, to implement appropriate countermeasures promptly.

- To address this limitation, our key idea is to combine the outputs of existing security solutions to build a probabilistic model (Bayesian network) that correlates multiple aspects of a 5G network and encapsulates the evolving nature of an APT attack over time. Consequently, an admin can derive security posture values for a given goal node (G_1 and G_2) from observed breaches or events (E_1 and E_2) to have a better understanding of the security status of the system. Furthermore, our approach enables the administrator to leverage previous data from time t_0 to t_n to predict the future security posture value (shown in blue) Δt time in advance, hence promoting proactive decision-making to avert attacks.

More specifically, we propose a 5G Security Posture Predictor (namely, *5GSPP*) using the results of existing security solutions (e.g., auditing and monitoring). First, we build an *event-state model* by integrating a *state model* derived from the auditing results and an *event model* based on the monitoring results to encapsulate attacker activities and their consequences. Second, we utilize this probabilistic event-state model to evaluate the security posture of a 5G network via Bayesian inference of the model. Finally, we employ the historical posture value derived from the event-state model to predict the future security posture value. We implement *5GSPP* utilizing free5GC [16] (a prominent project for 5G core deployment) alongside Kubernetes [17] (a leading container orchestrator), and validate its effectiveness through comprehensive experiments.

The key contributions of this paper are as follows.

- We propose a method to evaluate the security posture of 5G networks that can capture persistent attacks covering multiple aspects of 5G systems and the evolution of

threats over time.

- To accomplish this, we examine security controls from multiple aspects; we propose a semi-automatic method to determine the preconditions and postconditions of a security control breach; we introduce a novel *event-state model* by integrating findings from state auditing and event monitoring; we develop a customized algorithm to combine these results and evaluate the overall security posture of a 5G network.
- We integrate our solution with free5GC [18] and Kubernetes [17] and perform experiments to demonstrate the effectiveness of our event-state model in illustrating the impacts of attack progression, network scalability, and security measures on the overall security posture.
- We demonstrate various methods for predicting the future security posture of a 5G network by utilizing historical results from our event-state model along with historical events, thereby offering a configurable, fast, and precise prediction system.
- We create synthetic data and conduct experiments to show the effectiveness of different prediction methods by evaluating the accuracy and illustrating the impact of various conditions and goal nodes on the predicted value.

The preliminary version of this paper appeared in [19], which proposes the basic approach of evaluating the security posture of a 5G network. In this work, we significantly extend this approach to the proposed *5GSPP* system, while focusing more on the prediction aspects by proposing different approaches to predict the security posture of a 5G network and addressing several practical aspects of this system. Specifically, our major extensions are as follows: i) we design and implement a new system component for predicting future security posture to reduce the mitigation effort by early projecting the situations from the security posture value (Section V-D); ii) we explore different use cases of proposed approaches of posture prediction (Section VI); iii) we design and implement a new system component with a graphical user interface (GUI) to significantly improve the state model building process and assist the administrator in identifying the pre- and post-condition(s) of a security policy breach, which

helps to reduce the required manual efforts and improve the accuracy (Section V-B); iv) we update the related work section by incorporating prediction-related studies and a comparative table of existing works (Section IX). Finally, v) we evaluate both efficiency and accuracy of 5GSPP through a new set of experiments (Section VIII) which demonstrates significantly improved performance over our previous solution [19].

The rest of the paper is organized as follows. Section IV provides preliminaries. Section V describes our methodology. Section VI outlines various security posture prediction use cases. Section VII details its implementation. Section VIII presents experimental results. Section IX reviews the literature. Section X concludes the work.

IV. PRELIMINARIES

This section provides the necessary preliminaries.

A. Background on Multi-aspect 5G network and Its Challenges in Evaluating Security Posture

Figure 2 presents an overview of the multi-aspect 5G system, comprising three principal components: user, network, and infrastructure aspects. A user aspect pertains to the management of the 5G core network, encompassing User Equipment (UE), which may include both mobile and IoT devices. A network aspect includes equipment like RAN and 5G mobile core, as well as virtual network functions like AMF, UPF, SMF, and UDM [20]. An infrastructure aspect encompasses virtual resources (e.g., vSwitches, vRouters, vServers) generally situated on cloud infrastructure. We also summarize the acronyms used in this paper in Table IV.

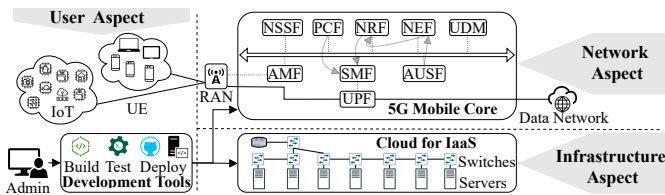


Fig. 2: An overview of the 5G network

TABLE IV: Acronyms used in this paper

Term	Description	Term	Description	Term	Description
3GPP	3rd Generation Partnership Project	5GC	5G Core	AMF	5G Access and Mobility Management Function
AUSF	5G Authentication Server Function	BN	Bayesian Network	CN	Core Network
EM	Event Model	ESM	Event-State Model	LTE	Long-Term Evolution
NEF	5G Network Exposure Function	NF	Network Functions	NRF	5G Network Repository Function
NSSF	5G Network Slice Selection Function	PB	Policy Breach	PCF	5G Policy Control Function
RAN	Radio Access Network	SM	State Model	SMF	5G Session Management Function
UDM	5G Unified Data Management	UE	User Equipments	UPF	5G User Plane Function

Challenges in Evaluating and Predicting Security Posture. The primary challenges to evaluating and predicting the security posture in 5G are:

- **Complexity of 5G network composition.** In contrast to pre-5G networks (e.g., LTE and 4G), 5G network components (e.g., CN, RAN, UE) are distributed across user, network, and infrastructure dimensions, allowing deployment and instantiation across third-party clouds and systems via SDN. The interplay between network functions, virtualized environments, and infrastructure owners complicates 5G networks. Due to this added complexity, evaluating a 5G network's security posture demands a deep understanding of the system.
- **Capturing multi-aspect attack progress with existing security tools.** Attackers may progress differently across multi-aspect 5G depending on their motivation, sophistication, objectives, and capabilities (which include skills, tools, and resources) [21]. The existing security solutions, however, are not enough to comprehend the connections between different breaches and to capture the attacker's progress to evaluate the system's overall security posture.
- **Interpreting and aggregating security solution outcomes.** Different 5G security solutions exist for auditing (e.g., [22]), monitoring (e.g., [10]), and alert reporting (e.g., [11]). Different (often inconsistent) security policies, result formats, and operating methods across those solutions may make correlation difficult when evaluating the system's overall security posture. Interpreting the binary results for evaluating the security posture of a 5G network is equally challenging.
- **Availability of training data and ground truth for prediction.** Predicting future security posture necessitates the training of a machine learning model with suitable training data and verifiable ground truth. Nevertheless, there is a lack of adequate training data and ground truth for the 5G scenario.

These challenges will be addressed in Section V.

B. Threat Model

This research addresses *in-scope threats* that can be identified by current auditing and monitoring techniques (e.g., [10], [11]) that impact the security status of a 5G network. In particular, the state model is capable of capturing any attacks, including zero-day attacks, that breach security control, regardless of the attack techniques. Whereas the event model captures attacker activity between breaches. We presume that the 5G network and infrastructure management systems (e.g., Kubernetes) can be relied upon for the integrity of API requests, event notifications, logs, and database records. The system may possess implementation flaws, misconfigurations, and vulnerabilities that could be exploited by malicious attackers to breach security controls. We assume that admins are interested in the security posture with respect to given attack goals (i.e., targeting critical assets). To identify pre- and post-breach conditions and privilege escalation nodes, we use expert intervention and the MITRE FiGHT [23] knowledgebase, which extends the generic MITRE ATT&CK [24]. The accuracy of predictions relies on the employed approaches (e.g., LSTM, DBN, etc.) and can be readily integrated with others.

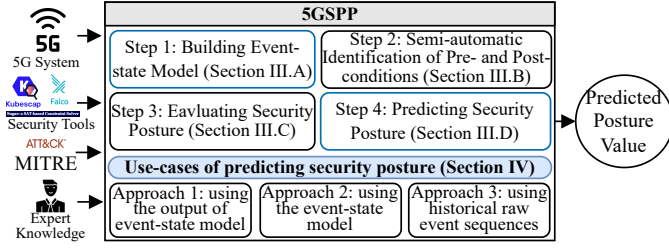


Fig. 3: Overview of our methodology and use-cases of predicting security posture

The *out-of-scope threats* include the attacks that do not violate the specified security controls, that are not captured by existing security solutions, and that may remove or tamper their own logged events. We also presume a loose synchronization of events across various components, ensuring that their order is accurately represented in Kubernetes logs and system logs. Even though this work can evaluate the effects of attacks on a system, its main objective is not to detect any specific attack, but to evaluate security postures for admins.

V. METHODOLOGY

This section presents the methodology of 5GSPP. Figure 3 shows an overview of our methodology, which contains four major steps. First, we employ the results of the security auditing and associated security controls to determine the pre-condition and post-condition of each security policy breach (as detailed in Section V-B). Second, we build an event-state model by fusing an event model derived from the monitoring result of 5G network events to capture an attacker's actions and a state model from the auditing results of 5G network states to identify compliance breaches by an attacker (as detailed in Section V-A). Third, we evaluate the security posture of a 5G network by transforming our event-state model into a probabilistic model represented as a Bayesian network (as detailed in Section V-C). Finally, we utilize historical events and security posture values produced by our event-state model over time to predict the future security posture value (as detailed in Section V-D). To achieve this, we employ three distinct approaches with three different kinds of input, which are historical posture values, historical event-state model with different conditions and historical raw events from security tools. We also demonstrate various use cases for each predictive approach in Section VI.

A. Building Event-State Model

To capture attacker activities and their impacts, an event-state model is created by merging the event model and state model, as depicted in Figure 4. We first construct a state model utilizing the previous auditing results and an event model based on the past security monitoring data to accomplish this. Subsequently, we apply our personalized fused algorithm to combine them.

Building State Model. We build the state model to encapsulate the attacker's actions that showcase the execution

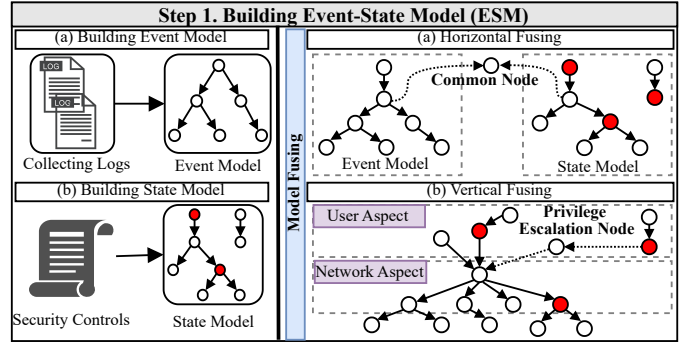


Fig. 4: Overview of Building Event-State Model

of malicious operations potentially leading to non-compliant system state. First, we collect the auditing results (e.g., compliant/non-compliant) over a period of time by applying existing security auditing tools (e.g., KubeScope [10]) and security controls (e.g., NIST [13]) covering different aspects of a 5G network. Then, we measure the frequency of both compliant and non-compliant results for each security control to calculate the probability of a non-compliant result. Next, we use the identified pre-condition (cause of the breach) and the post-condition (impact of the breach) for each non-compliant result using the semi-automatic approach described in Section V-B. Subsequently, for each non-compliant control, we construct a tiny graph by designating the pre-condition as the parent and the post-condition as the child, so enabling us to associate the activities of attackers in a non-compliant state with those pertaining to 5G. Finally, we combine all the tiny graphs to build a *state model* using model fusing (described in Section V-A).

Example 1. We use the same attack scenario as our motivating example to demonstrate how attackers compromise system states. Figure 5 demonstrates the state model construction process. Initially, an attacker performs an unauthorized logon that violates an AC-7: Authorized LogOn user-aspect control from NIST and enables the attacker to escalate to the network aspect. Next, s/he violates two more security controls (i.e., SI-4 and CM-7) through further malicious actions. To capture the effects of these malicious activities, a state model is built as follows. (i) We collect auditing results from three tools, KubeScope [10], Falco [11], and a custom tool based on formal verification [25]. (ii) We count the number of compliance (3) and non-compliance (2) for AC-7 and calculate the probability of non-compliance (0.40). (iii) For AC-7 policy breach we identify MITRE sub-technique Password Spraying as a pre-condition (cause of the breach), and Bypass User Account Control as a post-condition (a potential result of the breach) using our semi-automated tool (described in Section V-B). Afterwards, we connect the pre-condition as a parent and the post-condition as a child of the breach node of AC-7 control. Replicating these processes for other controls, we build the state model.

Building Event Model. To capture the attacker's activities supported by 5G network we build the event model. Initially,

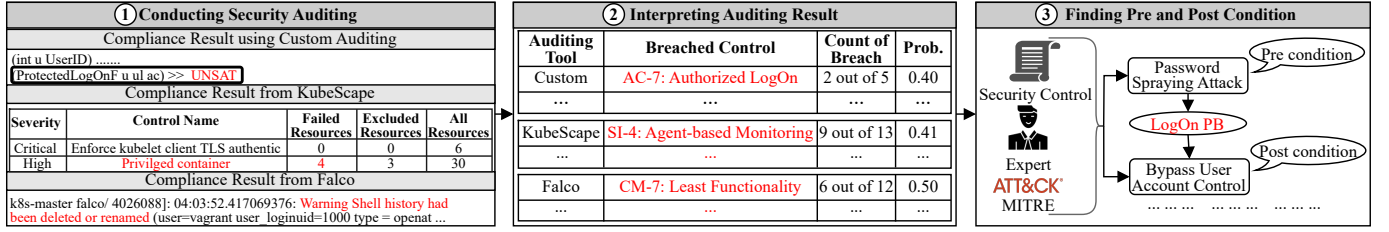


Fig. 5: Example of building a state model

we gather monitoring results (e.g., event logs) from several sources within a 5G network to cover its multiple aspects: system logs for users, auditing logs for infrastructure, and 5G core logs for networking. Then, we filter out any extraneous logs and trim the log data to extract the event-related information (e.g., timestamps and details of relevant events [26]). Subsequently, we combine all the events from different sources based on their timestamp, resulting in a single processed log. Moreover, we identify similar events (e.g., performing the same operations) that are named differently in different logs, and consider them as a single event type in our model. Then, to capture attacker activities (represented as a BN), we generate event sequences where each sequence ends as soon as any event is repeated to avoid cycles in our model (mainly because a BN is acyclic). Finally, we construct our *event model* (as further illustrated using the following example).

Example 2. Following the previous example, after gaining unauthorized access, an attacker may: access to server at the Kubernetes [17] level to alter the IP table to avoid detection, and query to SMF at the 5G level to get subscriber information from UDM. Figure 6 shows the steps to build the event model. (1) We collect and parse raw logs from Kubernetes and free5GC. (2) We remove the Log Level, Component, and Module columns, and the row with the message set report call: false as they do not provide event-related information. (3) We combine both logs, and sort them based on the *Timestamp*. (4) We rename both Created container UPF and Started container UPF messages as connect to UPF event, and identify the repetition of Query to SMF events for sequence generation without a cycle. (5) We obtain three event sequences. (6) We build an event model, learning nodes access server and Query to SMF and their transition probability (0.7) from sequences. Thus, the event model reveals attacker activities.

Model Fusing. To capture the overall impacts of the attacker's activities on the system, both models are fused in this step by performing two major steps as follows.

1) *Horizontal Fusing:* This step shows the combined effects of the attacker's activities causing a policy breach from the state model (SM) and in-between breach legitimate activities from the event model (EM). Initially, we identify all the common nodes between *EM* and *SM* caused by the impact of a privilege escalation resulting from a policy breach. Then, for each common node, we record and merge their parents and children from both models with their respective transition probabilities which yields a new model.

Example 3. In Figure 7a, we fuse the event model (on left) and state model (on right) using their common node, Access Server; which combines two different activities, Password-based Authentication representing attacker mimicking a legitimate user, and Bypass User Account Control representing a post-condition of a breach. Bypass User Account Control is also a common node which combine two breach-related activities LogOn Policy Breach (PB) and Remote Access PB within the state-model, using the same procedure. Dashed lines represent the model's instance prior to fusion, whereas solid lines denote its instance after to fusion.

2) *Vertical Fusing:* This step shows the impacts of a breach among different aspects (e.g., user, network) of a 5G network. First, we identify the privilege escalation (post-condition) that occurs as a result of a breach (explained in Section V-A). Then, using the MITRE FiGHT [23] and ATT&CK [24] framework, we identify the common impact of this privilege escalation. This impact could be a legitimate event from the EM or another breach of the system state from the SM. Furthermore, an expert uses their knowledge of attack and mitigation to confirm the decision. At last, we link the affected node with the privilege escalation node to demonstrate progress of the attacker across multiple aspects.

Example 4. As shown in Figure 7b, vertical fusing connects the user aspect and network aspect of the system where an attacker uses two breaches LogOn PB and Remote Access PB to gain the same privilege of Bypass User Account Control. Expert identify that Bypass User Account Control can be used for both Credential Access (MITRE TA0006) and Lateral Movement (MITRE TA0008) which impact is the Access Server in EM. So, we link Bypass User Account Control from the user aspect of the state model (on top), to Access Server from the network aspect of the event model (on bottom). This shows the attack progress from the user aspect to the network aspect and allows attackers to execute the Query to SMF operation.

B. Semi-automatic Identification of Pre- and Post-conditions of Breached Security Control

We identify the pre-condition (cause of the breach) and the post-condition (impact of the breach) for each non-compliant result of security results using the description of the control and MITRE FiGHT [23] and ATT&CK framework [24], as follows.

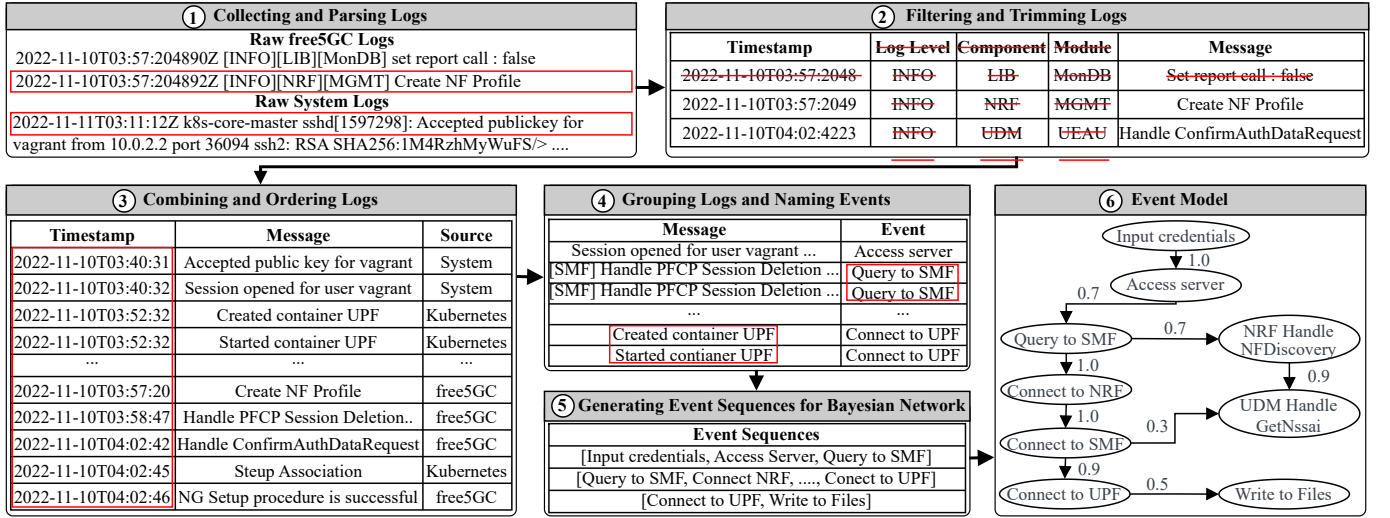


Fig. 6: Example of building an event model

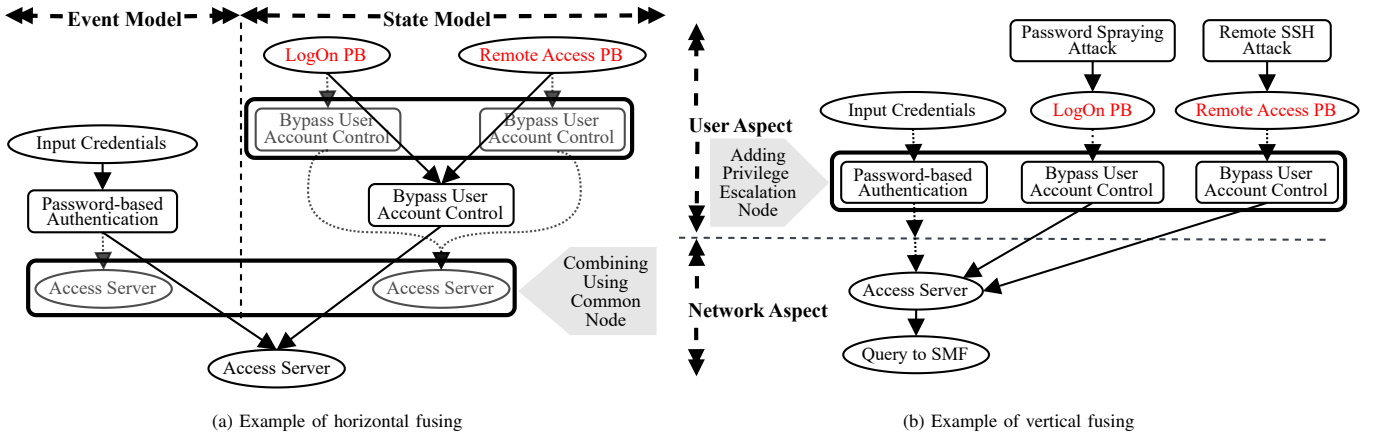


Fig. 7: Examples of fusing event model with state model into event-state model

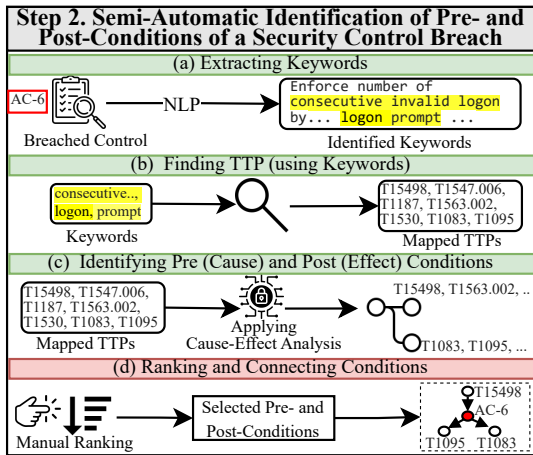


Fig. 8: Overview of Semi-automatic Identification of Pre- and Post-conditions of Breached Security Control

Methodology on Building State-model. To automate the process of building the event-state model and reduce the dependency on an expert in identifying pre- and post-conditions of a breach we build a semi-automatic approach for generating

the state-model. To do this, first, we summarize the description of breached security control and extract keywords from it. Second, we use the extracted keywords to search for relevant tactics and techniques from MITRE. Third, we differentiate the pre-condition and post-condition by analyzing the matched tactics and techniques using cause-effect analysis. Finally, we take admin feedback to fine-tune the selection. Figure 8 shows the overview of the whole process.

User-interface on Guiding Admin to Build the State Model.

An excerpt of the graphical user interface (GUI) that has been built as a web application to present the graphical view of the designated pre-condition and post-condition is illustrated in Figure 9. On the top you will be able to view all the breached security control in an expandable table row-view. If you click on one of the breached security control it will show you description and discussion extracted from NIST-SP-800 [13] and also display the relevant pre-condition and post-condition that has been extracted automatically (discussed in Section V-B). The user interface will also display a description of the data extraction process as a label for each stage. The GUI also gives the user the option to enter their own pre- and

State Model Viewer of 5GSPP

Breached security control(s)

This list of controls results from the existing auditing (e.g., KubeScape, Nmap, OpenVAS, etc.) solutions that are reported as breached. Please click on a control to view the details

ID	Title	Description
▼ AC-7	LogOn Policy Breach (PB)	Enforce a limit of consecutive invalid logon attempts by a user during an organization-defined time period ... see more
▼ AC-17	Remote Access	Virtual private networks can be used to protect ... see more

Extracted Keywords Top 5 Top 10

This list of keywords are automatically extracted using the help of Natural Language Processing (NLP) tools (e.g., spacy, textrank, etc.) from the description of the specific control from NIST-800-53. The ranking of the keywords is also provided by the NLP tools.

logon, invalid logon, system administrator, unauthorized logon, systems

Suggested pre-condition(s) of the breached control More pre-conditions Add yours

Automatically generated via NLP tools using the above extracted keywords and MITRE ATT&CK framework. Click on the "More pre-conditions" button to view other suggested pre-conditions. Click on the "Add Yours" button to input your own pre-conditions.

Selected	ID	Title	Description
☒	T1134	Access Token Manipulation	Adversaries may modify access tokens to operate under a different user or system security context to perform actions and bypass access controls

Detail Description: Adversaries may modify access tokens to operate under a different ...[see more](#)

Suggested post-condition(s) of the breached control More post-conditions Add yours

Selected	ID	Title	Description
☐	T1556	Modify Authentication Process	Adversaries may modify authentication ... see more

Save the Model
Cancel

Generated State Model

Selected breached control

Full model

Export

Fig. 9: Snapshot of the User Interface of the semi-automatic approach of building state model

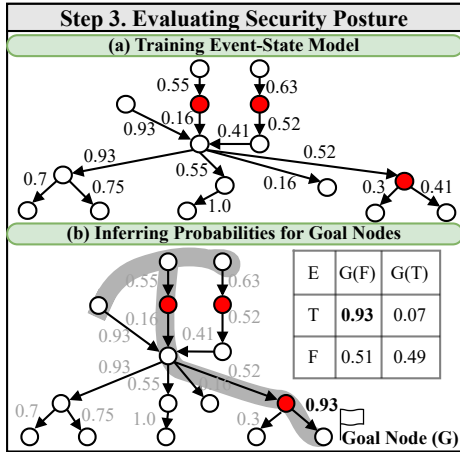


Fig. 10: Overview of Evaluating the Security Posture

post-conditions by clicking the Add Yours button if they believe the automatically extracted pre- and post-conditions are inappropriate for the chosen compromised security control. The user will also be able to see a graphical representation of the connection among the pre-condition, post-condition, and the breached security control (on the right side). Finally, the user can click on the Save Model button to save the model and export the graphical view in different formats (e.g., .jpg, .pdf, etc.).

C. Evaluating the Security Posture

Evaluating the security posture is performed using the following two steps depicted in Figure 10.

Training Event-State Model. Initially, we employ the fused model from the preceding stage to build the structure of the Bayesian Network (BN). We utilize past log data from both

monitoring events and auditing states of a 5G network to ascertain its parameters, similar to training a machine learning model. In the BN model, each incoming edge to a node shows the probability of an action. Nodes are represented as follows: (i) white nodes represent normal events, (ii) red nodes represent a breach state, and (iii) rectangular nodes represent privilege escalation, as depicted in Figure 11.

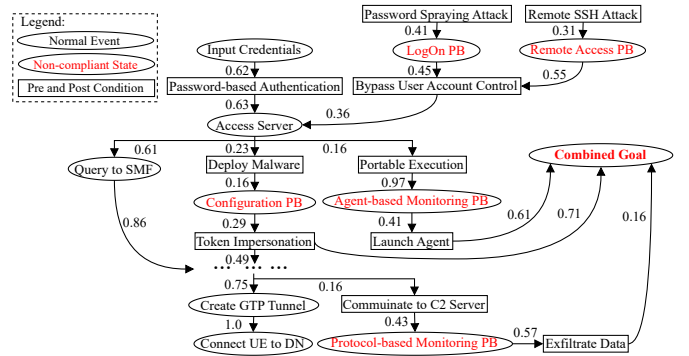


Fig. 11: Evaluating security posture based on the event-state model

Inferring Probabilities for Goal Nodes. This step use Bayesian inference to calculate the conditional probability of the goal node from the structured even-state model using the system's observed conditions (node values: occurred or not) and the user-defined target node where a *goal node* is an event or state of a system that are chosen by the admins based on the security requirements of the organization. We use Bayesian inference to calculate the conditional probability of the goal node. This probability measures an attacker's ability to reach the goal node, so from an administrator's perspective, security posture is the value that complements it. Choosing multiple

goal nodes is also possible in the real world. To achieve this, we propose the use of a dummy node that can be connected to all the goal nodes. Then, we re-build the model to incorporate the dummy node and recalculate the conditional probability for each node. Using the same approach, we can infer the value for this combined node which represents the combined security posture for all those goal nodes.

Example 5. Figure 11 shows an excerpt of our event-state model where we evaluate the security posture for the user-defined goal: Exfiltrate Data. Admin set the value of Access Server, Deploy Malware, and Configuration PB as occurred as observed conditions. Then Bayesian inference method on the event-state model yields a conditional probability of 0.57 and taking the complement, we get the security posture value of 0.43 for Exfiltrate Data. To exemplify multiple goal nodes, a new node is created and labeled as Combined Goal which connects three different goal nodes (Token Impersonation, Launch Agent, and Exfiltrate Data). After rebuilding the model using the same observed conditions, the new security posture value is 0.16 because there are now more paths to the combined goal, thereby decreasing the security posture value.

D. Predicting Security Posture

In this step, we use the historical data generated by our event-state model to predict the future security posture. To achieve this, we employ three distinct approaches. First, we use the output of our event-state model (i.e., posture value) over time to predict the future security posture value. We consider this as a time-series prediction problem where the historical security posture value over time is used as input. Second, the event-state model itself is used as an input as time-series data to predict future posture values based on various scenarios. Finally, we use the raw event sequences as input to predict future events, and then we apply our event-state model to calculate the future security posture value. Figure 12 shows an overview of different security posture prediction approaches.

Prediction Approach 1: Predicting Using the Output of the Event-state Model (i.e., Historical Posture Value). This approach utilizes the security posture value that the event-state model outputs to obtain a quick prediction without needing any further user input. We first split our data into training (70%) and testing (30%) data. Subsequently, the data undergoes preprocessing procedures such as normalization, data organization utilizing the sliding window (the number of events to consider when predicting subsequent events in a sequence), etc. After that, we use the training data to train an LSTM model [27], and the testing data to assess the model's accuracy. Lastly, we use the prediction method of the LSTM model to predict the future security posture value. This approach predicts future posture values based on data trends. Because of this, this method will work well when patterns of security posture change very slowly or when there is a pattern over time. Both long-term (e.g., security posture pattern in

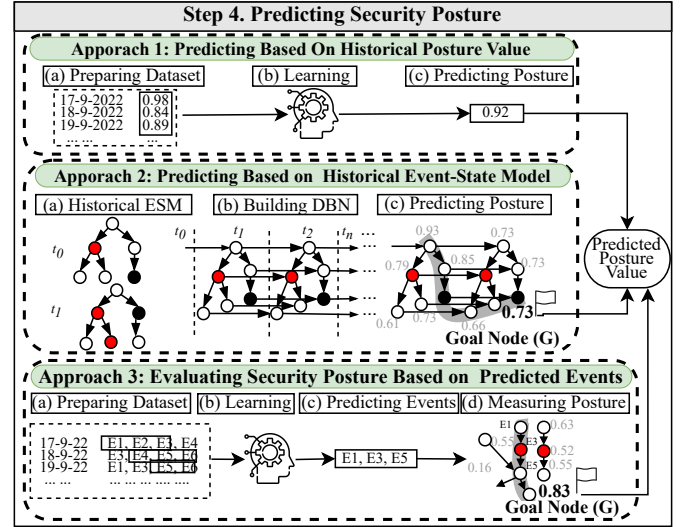


Fig. 12: Overview of Security Posture Prediction (ESM: Event-State Model, DBN: Dynamic Bayesian Network)

one season or year will be followed by another) and short-term (e.g., predictions for the next hour or day) predictions will benefit from it. This strategy won't be effective, though, if there are random fluctuations in the security posture value as a result of various attack types (such as zero-day attacks).

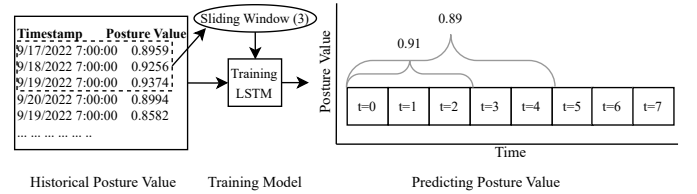


Fig. 13: Example of approach 1: predicting based on historical posture value outputted by event-state model

Example 6. Figure 13 illustrates an example of predicting future posture values solely based on previous posture values derived from the event-state model. On the left, sample historical data is shown, comprising solely the timestamp and posture value of a goal node (e.g., a combined goal node). First, we use two different sliding windows of 3 and 5 to pre-process our input data to make it sequential and suitable for the LSTM model. Using a sliding window of three to process the data means we use the last three days' security posture value 0.8959, 0.9256, 0.9374 to predict the fourth day's value as depicted on the right side of the figure. The predicted future posture value of the goal node becomes 0.91. Similarly, when we use a sliding window of size five, the predicted posture value becomes 0.89.

Prediction Approach 2: Predicting Using Historical Event-state Model. In this approach, different event-state models (i.e., multiple Bayesian networks) over time are utilized to incorporate more information (e.g., different node names, transitional probabilities from one node to another, etc.) into

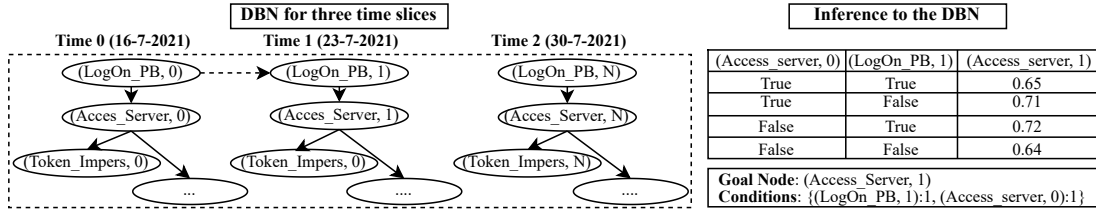


Fig. 14: Example of approach 2: predicting based on the historical event-state model

the prediction steps and to get better accuracy than the previous approach. We assume that we have stored the event-state model over time. Initially, we transform the node names of each event-state model by linking them with timestamps to integrate time series into the data. Then, we use a dynamic Bayesian network (DBN) [28] to integrate multiple event-state models and construct the DBN's structure. Following that, we train the DBN either using historical data from that time period or directly utilizing the conditional probabilities from historical event state models. Finally, we use the inference capability of the DBN, given the target node (i.e., goal node) and observed conditions, to determine the future posture value. The unique benefit of this approach is that the admin can accommodate customized conditions (e.g., an event or breach happened or not) and see their impacts on the overall 5G network by observing the posture value. The integration of numerous event-state models enables the generation of predictions with greater accuracy. This method allows for the experimentation of different conditions to identify the most efficient attack vector and attack path, facilitating the proactive mitigation. Nevertheless, this approach lacks the ability to predict an attack that does not breach any security policy. Also, it is dependent on the input of the administrators where they are required to input the conditions themselves to know the security posture value for a goal node.

Example 7. Figure 14 shows an example of predicting future posture value using a historical event state model. On the left, it depicts a few sample event-state models over time which are combined using the DBN. First, the nodes of each event-state model are renamed using the timestamp number where 0 indicates the event-state model build on 16th August, 1 indicates the event-state model build on 23rd August, and so on. The combined model is depicted using a rectangular box where dotted arrows are used to show the connection among the models. On the right-hand side, it shows the inference capacity of the DBN. Given the goal node is (Access_Server, 1) and observed conditions are (LogOn_PB, 0): True, (LogOn_PB, 1): True, (Access_server, 0): True, DBN inference capability predicted posture value for (Access_Server, 1) is 0.89. The table on the right-hand side shows different combinations of observed conditions where True means the condition has been observed and False means the condition has not been observed. These different combinations give us an idea of different paths to the goal node.

Prediction Approach 3: Predicting Using the Historical Raw Event Sequences. This approach explains the cause

for changes in the security posture value by first predicting future events using the raw event sequences and subsequently calculating the future security posture value using an event-state model based on the predicted events. We use the previous raw event sequence(s) across time as an input to train an LSTM model to predict future events or event sequences. We begin by dividing our historical raw event sequences into training and testing sets. Then, we process the data using a sliding window. Training data is used to train the LSTM model, while testing data is used to evaluate the model's accuracy. Next, we utilize the predictive abilities of the LSTM model to predict future events. Finally, we use the predicted event or event sequence as an input to our event-state model, which provides a security posture value. The unique advantages of this approach are that we can have an idea of the probable future event that is predicted based on past historical events, allowing us to prepare for the impact or take preventative measures against the event or event sequences. Following the prediction of the event can also complement approach 2, where we can use the DBN to investigate the overall effect of the particular event or events on the 5G network. Furthermore, it can also be used to predict the possible actions of an attacker as it depicts future events or event sequences. Nonetheless, as a result of the added event prediction, this method may incur additional overhead.

Example 8. Figure 15 shows an example of approach 3, where on the left, it shows multiple event sequences over time, each comprised of one or multiple raw events. First, we divide this historical raw events data into training (70%) and testing (30%) data. We utilize a sliding window with a value of three, meaning that the previous three event sequences are used to predict the next event or event sequence. The training data is then fed into an LSTM model and trained. Then we test the model accuracy with the test data. After that, we predict the future event sequences using the predict method of LSTM which results: Bypass_Auth, , Access_Server, Deploy_Malware. Finally, the predicted event sequences and their combinations are used as observed conditions to calculate the security posture value for the combined goal node using our event-state model. The combination of Bypass_Auth, Access_Server, and Deploy_Malware produces a security posture value of 0.91, whereas the combination of LogOn_PB, Bypass_Auth, and Access_Server gives us a security posture value of 0.76. Now, we know that LogOn_PB, and Deploy_Malware are the events responsible for the drop in security posture value.

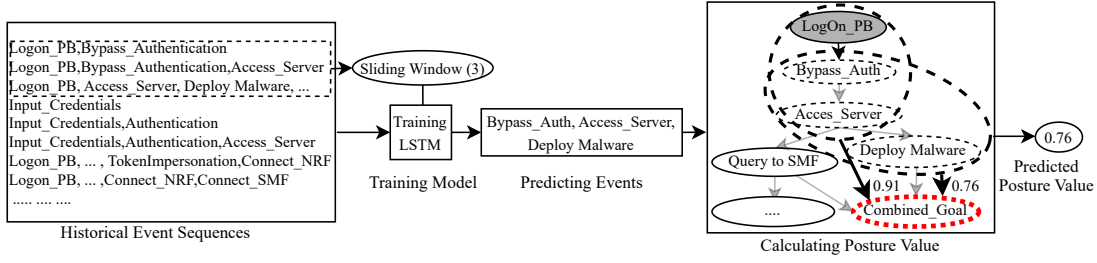


Fig. 15: Example of approach 3: Predicting based on the raw event sequences

VI. USE CASES ON PREDICTING SECURITY POSTURE

The use cases of the presented three approaches to predict security posture are detailed in this section.

Approach 1: Predicting Using the Output of the Event-state Model (i.e., Historical Posture Value). The main benefit of this approach is that it does not require any inputs to predict the security posture and requires very little time. For this reason, this approach is very useful for the higher management and decision-makers of the system, who have very little time and do not want to know the details of the system. All security professionals and executive leadership within a business can utilize this approach on a monthly basis to promptly evaluate the security status and engage in meaningful conversations with senior executives regarding its consequences. For example, as depicted in Figure 16, 5GSPP provides a graphical model of the whole system, and a senior executive (i.e., CEO, CSO, etc.) can click on any goal node (i.e., token impersonation, launch agent, etc.) and check the predicted security posture value without providing any user input and without the need for outside consultants to have a quick idea about the security posture value of different goal nodes. Moreover, the predictive nature based on the periodic data helps them to decide on different aspects. For example on the right-hand side of the figure 16 (bottom graph) shows the predicted security posture value over time, which dropped from time t2 to time t3. The management can see there is a monitoring policy breach, and they can decide to buy security monitoring tools. In another scenario (right-top) when the security posture value is predicted to be decreased each consecutively (from time t1, t2, and t3), they might decide to use an edge firewall to prevent a remote access policy breach. With a quick look, the management could have real, meaningful insights at their fingertips and start sharing them with the wider business. It can also be used to justify security spending during the budget process. Management can also use this solution to tailor their storytelling to the needs of functional stakeholders, including HR, legal, and marketing.

Approach 2: Predicting Using Historical Even-state Model.

Approach 2 mainly facilitates the simulation of different conditions (i.e., LogOn policy breach occurred) and checks their impacts on the security posture value. One such use case is depicted in Figure 17. Where the security admin suspects something but has not received any alerts yet from the monitoring tools. Now, the admin will simulate and provide certain conditions (e.g., observed LogOnPB) that might cause

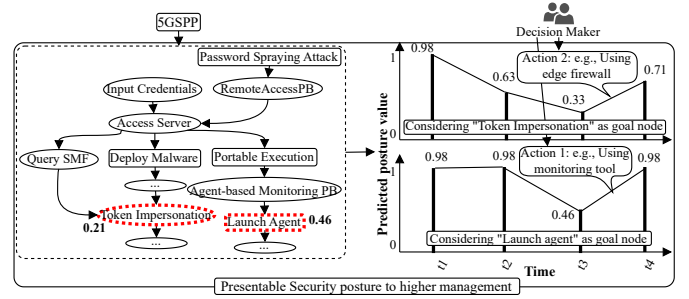


Fig. 16: Example use case of Approach 1

some alerts and check how goal nodes (e.g., Token impersonation, Launch agent, etc.) will be affected by these conditions to know which condition has more impact. The admin first triggered only LogOnPB and checked that the security posture value dropped for both goal nodes to 0.64 and 0.78 from 0.98 respectively (left side model). Then, they triggered both the LogOnPB and DeployMalware, and they found that these two conditions are responsible for a security posture drop of more than 50% (middle-model). Finally, they also checked the effects of modifying some snort rules as a defense mechanism, and they found that if they modified the snort rules, the security posture value improved to 0.89, and 0.96 respectively for TokenImp and Launch agent goal node (right side model). Then, they can prepare themselves for the mitigation in advance.

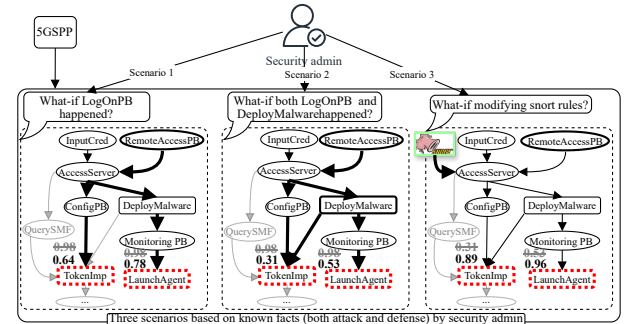


Fig. 17: Example use case of Approach 2

Approach 3: Predicting Using the Historical Raw Event Sequences. Approach 3 predicts the probable events to happen next. Figure 18 shows an example use-case of this approach. Upon receiving an alert from the monitoring tool that the LogOnPB is happening, the admin wanted to know what could happen next. 5GSPP predicts AccessServer,

PortableExecution as the next probable events and predicts the security posture value for different goal nodes. As shown on the right-hand side the security posture value drops from 0.98 to 0.86 when only RemoteAccessPB and Access_Server are used as the predicted events whereas it drops to 0.79 when RemoteAccessPB, Access_Server, and Portable_Execution are used as predicted events. This helps them to identify new conditions portable_Execution as a new event and helps them to take proactive measures and keep the security posture value high at all times.

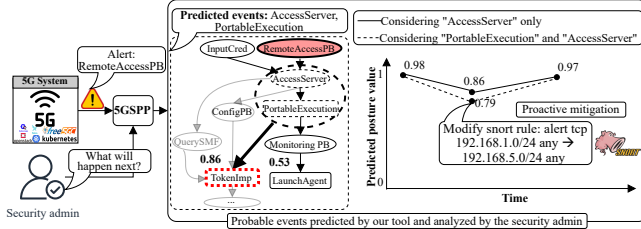


Fig. 18: Example use case of Approach 3

VII. IMPLEMENTATION

This section covers the implementation and integration specifics of 5GSPP.

A. Testbed Implementation of 5G

To implement our 5G testbed, we use Towards5G-helm [18] to automate the deployment of Free5GC [16] (version 3.2.1) on top of Kubernetes [17] (version 1.22). The RAN simulation is carried out by collaborating with UERANSIM¹ for testing of 5G core functions. We also utilize the existing auditing tools, Falco [11], KubeScope [10], and a Sugar [25] based custom tool, to monitor and audit the security of a 5G network. We utilize KubeScope (v2.0.183) [10] to identify potential vulnerabilities and misconfigurations within a Kubernetes cluster. As KubeScope does not focus on run-time detection, we use Falco (v0.33.1) [11] for detecting potential run-time security breaches in Kubernetes. However, Falco does not perform audits on activities within 5G network functions; hence, we employ customized detection criteria and a first-order logic-based tool named Sugar [25] to identify breaches in 5G network functions.

Challenges. We encounter a memory limit issue in Kubescope, and reconfigure the kube-apiserver.yaml file and set `--audit-webhook-mode` to blocking so that the Kubernetes API server does not send a response for each event. We also use `journactl` to dynamically retrieve all Falco alerts to address the issue of varying time precision among tools.

B. 5GSPP Implementation and Integration to free5GC

The system architecture of 5GSPP is shown in Figure 19. First, the *model builder* module is responsible for building

both the event and state model. 5GSPP initially aggregates logs from several components, such as Kubernetes, Falco, and KubeScope, and subsequently pre-processes them to identify event sequences from the logs. These event sequences are converted to the input (in .text) to PGMPY [29] (v0.1.19), a Python library used for building models and learning with a Bayesian network. Secondly, to build the state model, we first implement a breach analyzer and condition identifier to analyze the security breach and create a breach table. The model fuser module is implemented based on a customized algorithm (Algorithm 1 in [19]). Third, the security posture evaluator module uses different classes (e.g., Max-Product Linear Programming (MPLP) and variable elimination) from PGMPY to evaluate the security posture. Finally, the security posture predictor module uses TensorFlow based Keras library to implement the LSTM model and PGMPY to implement the DBN. We also use graphviz [30] to visualize the models, networkX [31] to implement various graph operations (e.g., `add_node`, `find_cycle`).

Challenges. The mapping of breached security control with pre- and post-conditions is a tedious and time-intensive endeavour; thus, we employ NLP tools (e.g., BERT [32], NER [33]) to address this issue. Graphviz's default layouts limit the user's ability to analyze, and arrange nodes and edges aesthetically; therefore, we employ pydotplus [34] coupled with Graphviz to solve this issue. We encounter a memory overflow problem when deploying DBN with above 15 nodes, which has been addressed by restricting the event-state model to a fixed size of 10 for experimental purposes.

VIII. EXPERIMENTS

This section describes the experimental setup and results.

A. Experimental Setup

The testbed is established in an OpenStack [35] environment, comprising one master node and two worker nodes on a server operating Ubuntu 20.04. In addition to our testbed data, we produce synthetic data in a simulated environment that creates 25,000 event sequences including 25 unique event types including breaches of specific security controls from NIST [13] (AC-7: Logon Policy Breach (PB), AC-17: Remote access PB, AC-8: System use notification PB, CM-7: Configuration PB, SI-4 (2): Agent-based monitoring PB, SI-4 (7): Rule-based monitoring PB). To synthetically construct event sequences, we adhere to the dependencies among event types identified in our testbed and allocate various event types to attackers in a random but realistic manner. We generate attackers with different capabilities (which is the number of events (both regular and attack events) in a sequence that an attacker is capable of completing) using an exponential distribution (where the majority of attackers possess an initial capability and comparatively fewer attackers with higher capabilities) and assign generated events accordingly. We also divide each event sequence into three attack stages based on the MITRE ATT&CK framework,

¹<https://github.com/aligungr/UERANSIM>

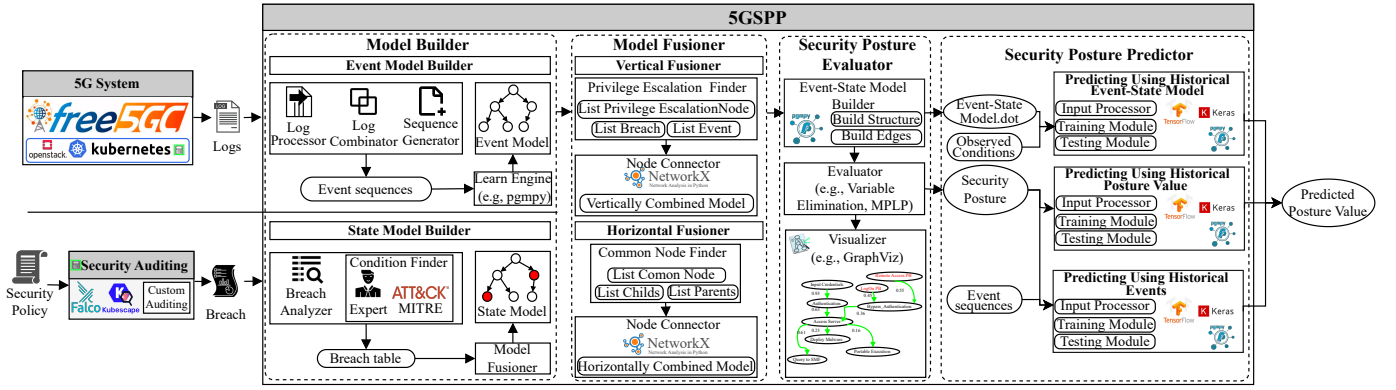


Fig. 19: System architecture of 5GSPP

with each stage showing the attacker's attack progression (in Table V). We conduct each experiment 1,000 times and calculate the average value.

TABLE V: Statistics of our dataset

Attack Stage	MITRE Tactics	No. of Seq.	Attacker Level	Sequence Length	No. of Seq.
1	Reconnaissance to Initial Access stages	9,290	1	1 to 2	17,242
2	Execution to Credential Access stages	5,938	2	3 to 4	5,518
3	Discovery to Exfiltration stages	10,556	3	5 to 6	1,737
	Total	25,784	4	more than 6	1,287

B. Experimental Results on Evaluating the Security Posture

Here, we provide our experimental results for evaluating the security posture.

Effects of Attack Progress. The objective of our first set of experiments is to accurately determine the overall security posture as the attack progresses. Experiment parameters is composed of various attack stages and goal nodes. To more accurately interpret the security posture values that are derived from 5GSPP, we also quantify the attacker's success ratio (ASR), which is the percentage of successful attackers that reach the goal node during an attack stage. Figures 20a, 20b, 20c, and 20d show the relationship between the attack stages and the overall security posture value for different goal nodes: combined goal, exfiltrate data, launch agent, and token impersonation, respectively. We find that the attacker success ratio and the security posture values are approximately inversely related across all goal nodes. This is due to the fact that a higher security posture value suggests a better-secured system, and vice versa. On the Figure 20 shows this, as an attack progresses from one stage to the next, the overall security posture value decreases. Moreover, the average security posture value reduces more when the goal node is closer to the attacker (e.g., for token impersonation as the goal node, the average posture value decreases by 29% (0.93 to 0.66 in Figure 20d) whereas, for exfiltrate data as target node, it decreases by just 1% (0.99 to 0.98 in Figure 20b) from attack Stage 1 to attack Stage 2. Consequently, we can ascertain that, the attack stage significantly influences the overall security posture.

Effects of Scaling Network Size (of 5G Core network). This second set of experiments examines how network size affects security posture. We construct several 5G core networks by altering the number of Network Functions (e.g., UPF and AMF) from 1 to 20 [36] and produce synthetic data (as detailed in Section VIII-A). Subsequently, we compute average security posture value, event-state model size (total number of nodes and edges), time to build the model and time to evaluate it for different network sizes. Figure 21a depicts the linear increase in model size due to network scaling which remains reasonably small compared to the size of other related graphs (e.g., attack graph [37], which can exceed 10,000 nodes). Here, NF1 represents a user plane function, and NF2 represents a control plane function. Figure 21b illustrates that both model building time and prediction duration increase linearly with network scaling. Nevertheless, for up to 15 NFs, the time required to build the model is under two seconds, indicating that our approach demonstrates efficiency up to a specific degree of network scaling. We also notice a large increase in the build time for 20 NFs, which may be caused by multiple NFs connecting to other NFs, increasing the number of edges. Figure 21c shows that security posture value does not change significantly with network size, as it does not necessarily increase security control breaches. The red line in the figure shows an enlarged NF2_EXfiltrate on the right y-axis with a different scale. The security change is negligible (0.02 when increasing NFs from 1 to 20).

Effects of Attacker Capabilities. The third set of experiments shows how attacker capabilities affect the overall security posture. Figure 21d shows that as attacker capability increases, security posture decreases. The average security posture value for an attacker with capability *Level 1* (the lowest, defined in Table V) is 0.98, whereas, for an attacker with capability *Level 4* (the highest), is 0.78 for the combined goal node. Additionally, as most attackers with initial capability level (e.g., 1) can reach the goal node closer to the attacker (e.g., Token Impersonation) more easily, the security posture value is lower for that goal node than for the far-end goal node (e.g., Exfiltrate data). Nevertheless, as the combined goal node also takes into account all other goal nodes, its security posture value is situated between those of the other goal nodes.

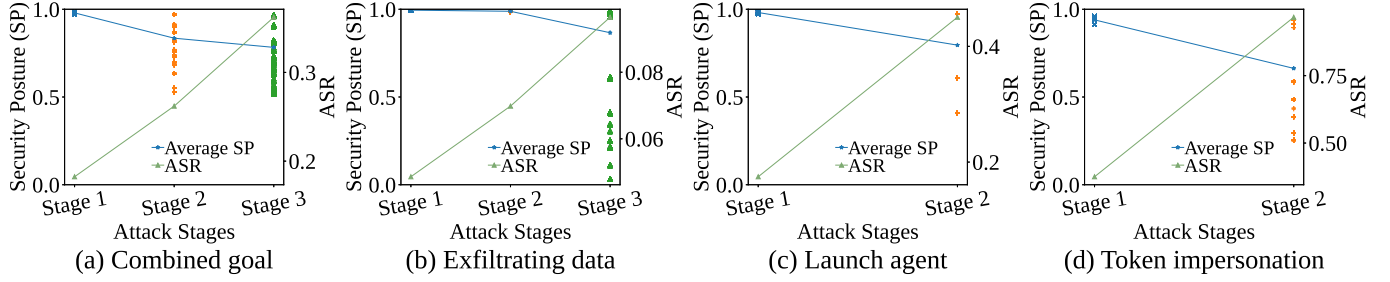


Fig. 20: Effects of attack progress for different goals; ASR: Attacker Success Ratio

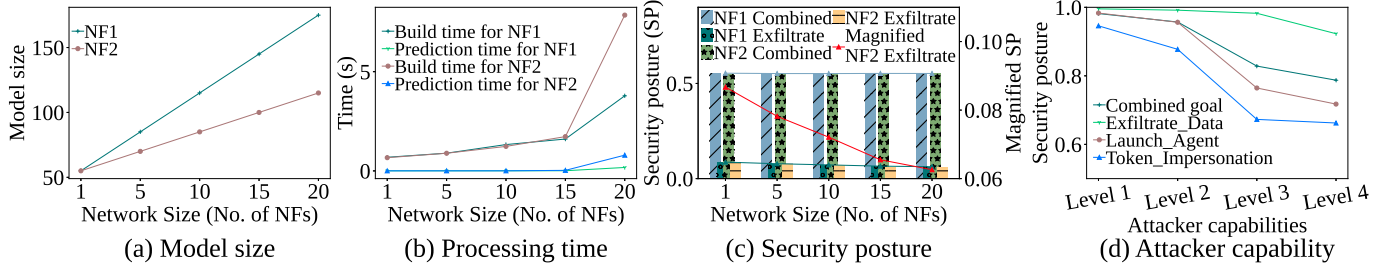


Fig. 21: Effects of scaling network size and Effects of attacker capabilities

Effects of Reducing Policy Breach by Implementing Security Solutions. The final set of experiments investigates how 5GSPP determines the overall impact on security posture following the implementation of security measures within a system. Initially, we compute the security posture value utilizing the attack scenario depicted in Figure 1, which encompasses six policy breaches without mitigating any of them. Subsequently, we mitigate individual security policy breaches (as a possible consequence of deployed security solutions) and recalculate the security posture value. Figure 22a shows that reducing security policy breaches leads to a reduced model size, with the greatest reduction occurring with more than two policy breaches. Both build time and prediction time also decrease, but we do not observe a significant change as the model is not large enough (in Figure 22b). The security posture value is at its lowest when no policy breach is addressed (without reduction), as shown in Figure 22c. It rises as security policy breaches are resolved. The red line illustrates a magnified version of `Exfiltrate Data` using the right y-axis with a different scale which shows an increase in security posture value. Figure 22d substantiates this trend with data from two separate attacks executed on our testbed: `LightBasin` (blue line) and a custom APT attack (orange line), both of which breach three security policies outlined in Table VI.

In both attacks, with all three policy breaches, the security posture is minimal, and it increases as the number of policy breaches reduces. The average security posture value increases for the custom APT attack from 0.62 with all three breaches (EIP, TSCF, and ACR) to 0.99 with no breaches.

C. Experimental Results on Predicting the Security Posture

Experimental results for security posture prediction are provided here.

Prediction Result Over Different Rate of Changes of Historical Posture Value for Approach 1. This experiment aims to demonstrate the accuracy of approach 1, wherein the historical security posture value changes over time in three formats: (i) *negligible change over time*, (ii) *varying rates of change exhibiting a pattern*, (iii) *varying rates of change without a discernible pattern* because the predicted posture value in our approach 1 is contingent upon the training data and its rate of change over time. First, we identify various rates of change. Subsequently, we generate historical security posture values within a specified range for each rate of change, ensuring that the average rate of change for the entire dataset is less than or equal to the identified rate of change. We employed seven values for rates of change: 0, 0.1, 0.3, 0.5, 0.7, 0.9, 1, where 0 signifies no modification in security posture value. Finally, we compute the root mean squared error (RMSE) and mean absolute error (MAE) at various rates of change to demonstrate accuracy.

The result shows that approach 1 exhibits optimal performance when there are negligible changes in the training data (i.e., security posture value). Figure 23a demonstrates the accuracy of the predicted posture value when the security posture remains constant over time (i.e., the rate of change is zero) and when it fluctuates minimally (i.e., the rate of change is between zero and 0.1). We also achieve better accuracy when the training data follows a pattern over time. Figure 23b shows the accuracy of the predicted posture value when the security posture value follows a pattern. We consider three different cases. In the first case, the security posture value follows a *single fixed length pattern* over time. For example, over time, the security posture value follows a single pattern of [0.98, 0.93, 0.88, 0.83, 0.78] where the pattern length is fixed, which is five. In the second case, we consider *multiple fixed-length patterns*. For exam-

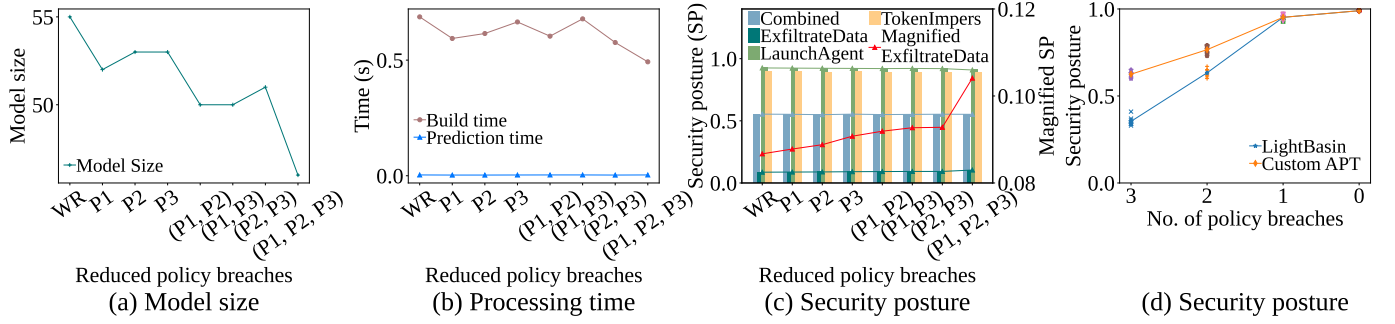


Fig. 22: Effects of reducing security policy breaches where WR: Without Reduction, P1: Reducing System Use Notification Policy Breach (PB), P2: Reducing Remote Access PB, P3: Reducing LogOn PB

TABLE VI: Implemented attacks' description

Attack	Attack Steps	Purpose	Breached Security Policy
Light Basin	Unexpected Inbound Connections (UIC)	To gain access to the system	AC-3 (Access Enforcement)
	Creating NodePort Services (CNS)	To expose containerized 5G Network Functions(NFs) to the Internet	CM-6 (Configuration Settings)
	Unexpected HTTP Requests to UDM (UHRU)	To exfiltrate subscribers' data from the UDM	SC-7 (Boundary Protection)
Custom APT	Exec Into Pods (EIP)	To execute malicious codes in pods	AC-3 (Access Enforcement)
	Tampering Shell Configuration Files (TSCF)	To weaken the system security	SI-7 (Software, Firmware, and Information Integrity)
	Attaching Cluster-admin Role (ACR)	To impersonate network admin.	AC-6 (Least Privilege)

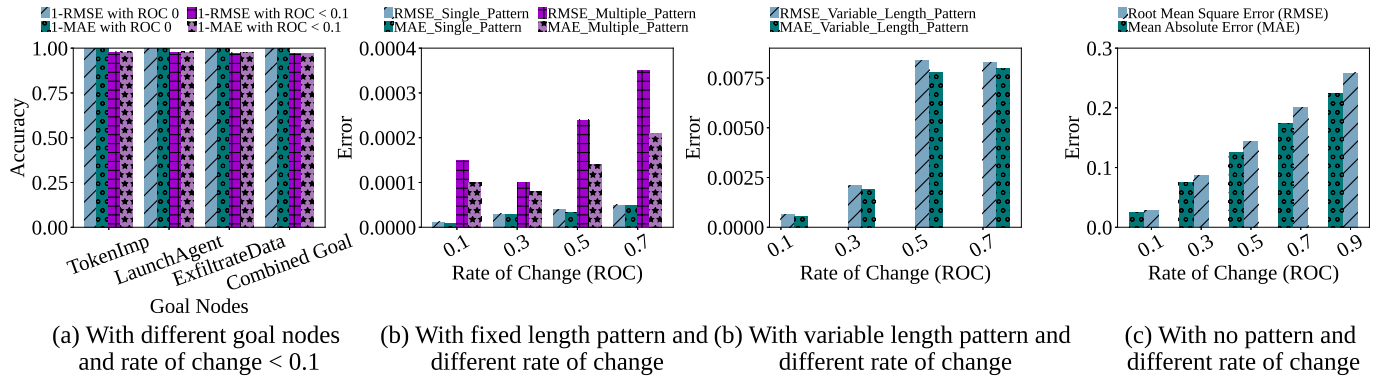


Fig. 23: Result of Approach 1 (predicting based on historical posture value)

ple, one pattern is $[0.98, 0.93, 0.88, 0.83, 0.78]$ and another is $[0.99, 0.97, 0.95, 0.93, 0.91]$ where both patterns have a fixed length of five. The third case is shown in Figure 23c, where the security posture follows a *pattern of different lengths*. For example, a security posture value with three patterns such as $[0.98, 0.94, 0.91]$, $[0.99, 0.97, 0.95, 0.93, 0.91]$, $[0.98, 0.97, 0.96, 0.95, 0.93, 0.92, 0.91]$ where all of them have different lengths (e.g., 3, 5, 7). We have observed that the model can predict the security posture value with an accuracy of nearly 100% or zero error in all three cases. Nevertheless, inaccuracy slightly escalates when the rate of change increases. Various rates of change in the training data, whether with a random pattern or without a pattern, have varying effects on the prediction accuracy, as illustrated in Figure 23d. We note that as the rate of change escalates, the error in the predictive model also increases. The greatest error (0.28) observed for the rate of change at 0.9 indicates that the absence of patterns in the data results in reduced prediction accuracy.

Prediction Result Over Different Numbers of Historical

Even-state Model for Approach 2. This set of experiments demonstrates the prediction accuracy based on the historical event-state model and various observed conditions, as Approach 2 is based on them.

First, we build different dynamic bayesian network (DBNs) with different numbers of historical event-state models. Subsequently, we evaluate the prediction accuracy for certain goal nodes under various conditions. All goal nodes exhibit a reduction in error as the number of event-state models increases, signifying improved accuracy with larger datasets, which is shown in Figure 24a. This is attributable to the increased availability of data, as each event-state model contributes additional data to the DBN. We also observe that TokenImp consistently demonstrates the lowest error across all goal nodes due to its reliance on fewer nodes.

Figures 24b and 24c illustrate the effects of varying conditions; the former depicts the influence of differing numbers of conditions, while the latter demonstrates the effects of diverse combinations of conditions. As shown, more observed conditions (e.g., C1C2C3C4 > C1C2) improve security posture

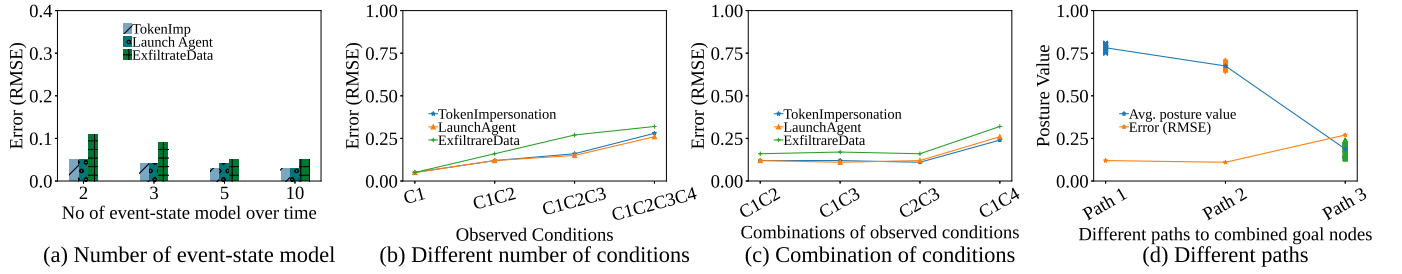


Fig. 24: Result of Approach 2 (predicting based on historical event-state model). P1 consists of C1=Access Server; C2=Deploy Malware; C3=Portable Execution; C4: Communicate to C2 Server

prediction because node conditional probabilities are changed based on observed conditions. However, the goal nodes that involve more conditions (i.e., Exfiltrate Data) produce less accurate results. This is due to the growth in the number of interconnections, which heightens the dependency on preceding nodes. The prediction results are also dependent on different combinations of observed conditions (i.e., C1C2, C1C3, C2C3, etc.). We observed that initial-level goal nodes (i.e., TokenImpersonation) are not affected much due to different combinations of observed conditions, while the deeper goal node (i.e., ExfiltrateData) shows less accurate results because only this goal node involves Condition 4 (i.e., Communicate to C2 Server) which has a direct impact on it. Figure 24d substantiates this, as combined goal nodes with three distinct paths (a combination of more than two observed conditions) exert varying effects on prediction accuracy. We observed that predicting using Path 3, which is via the Exfiltrate data (see Figure 11) node, has more security posture value drop and is less accurate because of different probable observed conditions on that path. Meanwhile, the path via token impersonation has more accuracy, and the security posture value is higher than that of the other two paths.

Prediction Result Over Different Data Sizes and Different Numbers Predicted Events for Approach 3. Figure 25a shows the accuracy of predicting events using the LSTM model for different data sizes where data size represents the number of event sequences over time. It has been observed that the accuracy of the prediction is linearly proportional to data size. With a data size of 20,000 repeated event sequences, we got the minimum error (maximum accuracy), which is 0.18. This indicates that if we have more repeated event sequences over time, our approach will perform well.

Figure 25b shows the accuracy of predicting the security posture value of different goal nodes for different numbers of predicted events. For different goal nodes, the accuracy of the predicted posture value does not differ much. Nonetheless, as the quantity of anticipated occurrences rose, the precision of the prediction diminished. This is to be expected as predicting additional events involves dealing with many probabilistic values.

Comparison of different approaches of predicting security posture value. The aim is to demonstrate the relative efficiency regarding accuracy and prediction time overhead for different approaches. To experiment with the accuracy of the

prediction approaches we use different rates of change over time and predict the security posture value. For each rate of change, we calculate the accuracy in terms of RMSE, and MAE. We observed that Approach 1 is less accurate than the other two approaches for an increased rate of change (i.e., 0.7) because this approach only uses the security posture value for prediction. Approach 2 shows better accuracy than Approach 1 and Approach 3, because Approach 2 is based on the event-state model, and it is predicted using the dynamic Bayesian network, which is very similar to our ground truth (value comes from the event-state model). Figure 25c shows the results.

Figure 25d shows the results of the prediction time overhead of different approaches. It can be used to analyze the efficiency of different prediction approaches in relation to the rate at which data changes. We observe that with a higher rate of change, Approach 1 takes less time than the other two. Also, for all rates of change, the prediction time is almost the same. For all three approaches, the prediction time is less than 0.1 seconds.

IX. RELATED WORKS

This section reviews the related literature.

Comparison of Existing Works. Table VII summarizes the comparison between existing works and 5GSPP. The first three columns enlist existing works, their objectives, and methodologies employed to predict the security posture. The third column specifies the work environment. The next three columns compare these works according to the different characteristics of evaluating the security posture. The characteristics of covering multiple aspects is checked when a solution uses data from multiple aspects (i.e., user, infrastructure, and network). When a solution addresses a multi-layer attack, we check the characteristics of the attack's progress. In addition, we use a dash symbol on the threat scope if the work does not apply to that category. In the last three columns of the table, we compare the presented works according to the different characteristics of predicting the security posture. Depending on the types of prediction approaches present in Section V-D (e.g., predicting posture value, predicting events, predicting based on conditions), we mark those types in the table.

In summary, 5GSPP mainly differs from the state-of-the-art works as follows. Firstly, it stands out as one of the few approaches that gather data from multiple aspects: user

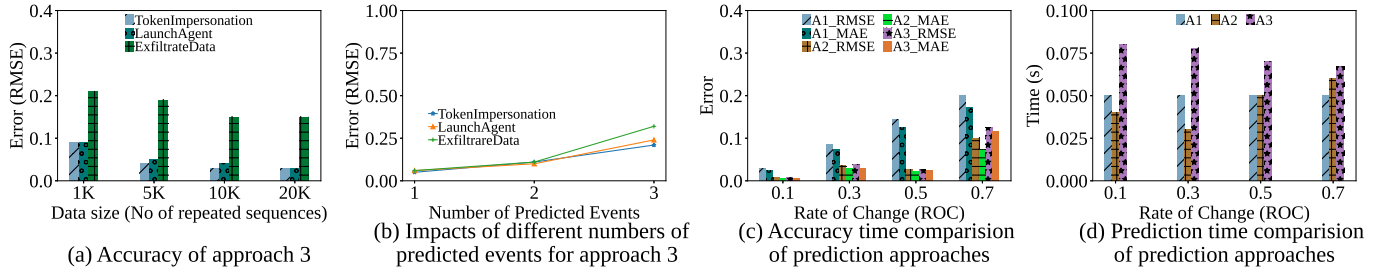


Fig. 25: Result of Approach 3 (predicting based on raw events) and comparison of all the three approaches (A1-A3)

TABLE VII: Comparing existing solutions with ours. (•), (◦), (–), and (NA) mean supported, partially supported, not supported, and not applicable, respectively.

Proposals	Objective		Methodology	Environment	Characteristics							
	Evaluating Security Posture	Predicting Security Posture			Evaluating				Predicting			
					Covering Multiple Aspects	Covering Attack Progress	Combining Event and State	Turning Binary Result	Customizable	Efficient	Accurate	Observation-based
5GSPE [2023] [19]	●	–	Bayesian Network	5G	●	●	●	●	NA	NA	NA	NA
Nie et al. [2022] [38]	●	–	Deployment based	5G	–	–	–	○	NA	NA	NA	NA
Zhao et al. [2021] [2]	●	–	Attack-graph	5G	○	–	–	○	NA	NA	NA	NA
Shafayat et al. [2022] [39]	●	–	Attack-graph based cross-layer approach	5G	○	–	–	○	NA	NA	NA	NA
Yang et al. [2019] [40]	●	●	Evidence inference rule	5G	–	–	–	●	○	–	○	–
Xu et al. [2024] [9]	●	●	Graph Convolutional Networks (GCNs)	Cloud	●	–	–	–	–	–	○	–
HereroScore [2023] [5]	○	–	Clustering and Heterogeneity	Cloud	○	–	–	–	NA	NA	NA	NA
Lei et al. [2022] [41]	●	–	Hybrid	Cloud	○	–	–	○	NA	NA	NA	NA
Frigault et al. [2007] [6]	●	–	Dynamic Bayesian Network	Network	●	○	●	○	NA	NA	NA	NA
Yang et al. [2023] [42]	●	●	Improved Hidden Markov Model (MMM)	Network	–	○	–	●	–	–	●	–
Li et al. [2023] [43]	–	●	Backpropagation Neural Network (BPNN)	Network	○	–	–	–	○	–	○	–
Zhao et al. [2023] [44]	●	●	BiLSTM, autoregression, and multi-scale CNN	IoT	○	–	–	○	–	–	●	–
Yang et al. [2023] [45]	○	●	Fused CNN-BiLSTM-Attention	IoT	○	–	–	●	–	–	●	–
Jiang et al. [2023] [46]	●	●	Minimum cut set method and manual prediction	Power	○	–	NA	–	○	–	–	○
Ma et al. [2023] [47]	○	●	Extreme learning machine	Power	–	–	–	○	NA	–	●	–
5GSPP	●	●	Bayesian Network based Event-State Model	●	●	●	●	●	●	●	●	●

data from Kubernetes, network data from free5GC, and infrastructure data from both. Secondly, 5GSPP is the only solution that incorporates all three aspects of the system to predict the security posture value. Finally, unlike alternative solutions, 5GSPP is the only solution to implement different approaches in predicting security posture based on different contexts, which provides a new perspective to this literature. Thus, we consider 5GSPP complementary to the other state-of-the-art solutions (that evaluate and predict the security posture). However, 5GSPP still has some limitations. It needs onetime adaptation effort when applied to a new system to train the model with the new data. 5GSPP mainly based on the historical data of the system and relying on existing works for collecting security incident-related data, such as monitoring

and auditing solutions.

Evaluating and Predicting Security Posture. Current research (e.g., [2], [39]) in 5G employs an attack graph-based approach to introduce a 5G-specific metric that utilizes cross-layer information exchanged among 5G network stakeholders, with an emphasis on the user perspective. Another study (e.g., [38]) also evaluates the security of commercial 5G deployment by considering only the user aspects (e.g., subscriber protection, user plane protection). Yang et al. [40] developed a security situation prediction model for 5G-oriented Multiple Input and Multiple Output (MIMO) systems based on the MIMO system theory. Nonetheless, each of them is confined to a specific aspect (e.g., user, network, or infrastructure) of a 5G network rather than the entire 5G network. There are a

few works (e.g., [48]) that quantify other aspects of 5G, but they do not evaluate its security posture.

Several research studies (e.g., [6], [7]) incorporate attack-graph analysis, wherein the probability and potential consequences of a particular threat are assessed through probabilistic models. Frigault et al. [6] integrate temporal elements through dynamic Bayesian networks and Wang et al. [7] concentrate on the causal linkages among vulnerabilities represented in attack graphs. Various studies (e.g., [5], [41], [49]) evaluate security metrics based on vulnerability or device behavior. Other works use GCN [9], optimization algorithm [43], Hidden Markov Model [42], decomposition-fusion algorithm [45], enhanced BiLSTM [44], minimum cut set method [46] and extreme learning machines [47] and others to enhance the network security posture prediction model for cloud, IoT and power system-specific studies. Nonetheless, the majority of the studies concentrate on predicting security situations rather than assessing the overall security posture. Also, none of them address 5G-specific challenges to capture the persistent threats with evolving impacts on multiple aspects of a 5G network.

Other Security Solutions in 5G. There are a few built-in tools (such as Kubescape [10] and Falco [11]) facilitate auditing and monitoring for Kubernetes-based 5G setups. Additionally, various auditing approaches are tailored for specific applications, including cloud auditing [50] and IoT auditing [51]. Other solutions in 5G (e.g., [52]–[54]) focus on various security objectives (e.g., attack and interference detection, log management, etc.) Furthermore, security information and event management (SIEM) solutions for 5G (e.g., [55], [56]) predominantly emphasize log management. However, none of those works focus on security posture evaluation.

X. CONCLUSION

In this paper, we introduced 5GSPP, a novel framework for evaluating and predicting the security posture of 5G networks while considering the 5G-specific security threats (e.g., persistent threats with evolving impacts from multiple aspects of a 5G network). In particular, we first built an event-state model by combining an event model (built using monitoring results) and a state model (built using auditing results). Subsequently, we evaluated the security posture of a 5G network utilizing Bayesian inference of the model. Finally, we use different prediction approaches to predict the future security posture based on historical data. We implemented and incorporated our solution for free5GC and Kubernetes, demonstrating its efficacy through comprehensive experiments.

Limitations and Future Work. We intend to apply the security posture prediction capability in various security applications (e.g., risk assessment). Also, we plan to apply our work to other systems (e.g., clouds, IoT).

REFERENCES

- [1] 5G Americas, “Global 5G Connections Reach Nearly Two Billion.” <https://www.5gamericas.org/global-5g-connections-reaches-nearly-two-billion/>, Dec 2024. (accessed: 24 Dec 2024).
- [2] L. Zhao, M. S. Oshman, M. Zhang, F. F. Moghaddam, S. Chander, and M. Pourzandi, “Towards 5G-ready security metrics,” in *ICC 2021-IEEE International Conference on Communications*, pp. 1–6, IEEE, 2021.
- [3] E. Yocam, A. Gawanmeh, A. Alomari, and W. Mansoor, “5G mobile networks: reviewing security control correctness for mischievous activity,” *SN Applied Sciences*, vol. 4, no. 11, pp. 1–17, 2022.
- [4] O. Duman, A. Tabiban, L. Wang, and M. Debbabi, “Measuring and Improving the Security Posture of IEC 61850 Substations against Supply Chain Attacks,” *IEEE Transactions on Instrumentation and Measurement*, 2024.
- [5] C. Fang, N. Nazari, B. Omid, H. Wang, A. Puri, M. Arora, S. Rafatirad, H. Homayoun, and K. N. Khasawneh, “HeteroScore: Evaluating and mitigating cloud security threats brought by heterogeneity,” in *The Network and Distributed System Security Symposium (NDSS)*, 2023.
- [6] M. Frigault, L. Wang, A. Singhal, and S. Jajodia, “Measuring network security using dynamic Bayesian network,” in *Proceedings of the 4th ACM workshop on Quality of protection*, pp. 23–30, 2008.
- [7] L. Wang, A. Singhal, and S. Jajodia, “Measuring the overall security of network configurations using attack graphs,” in *IFIP Annual Conference on Data and Applications Security and Privacy*, Springer, 2007.
- [8] M. Zhang, L. Wang, S. Jajodia, A. Singhal, and M. Albanese, “Network diversity: a security metric for evaluating the resilience of networks against zero-day attacks,” *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 5, pp. 1071–1086, 2016.
- [9] M. Xu, S. Liu, and X. Li, “Network security situation assessment and prediction method based on multimodal transformation in edge computing,” *Computer Communications*, vol. 215, pp. 103–111, 2024.
- [10] “Kubescape.” <https://github.com/kubescape/kubescape>, Dec. 2022.
- [11] Falco, “Falco.” <https://github.com/falcosecurity/falco>, Dec. 2022.
- [12] “LightBasin: A roaming threat to telecommunications companies.” <https://www.crowdstrike.com/blog/an-analysis-of-lightbasin-telecommunications-attacks/>, Dec. 2022.
- [13] NIST, “SP 800-53 Rev. 5, Security and Privacy Controls for Information Systems and Organizations.” <https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final>, Dec. 2022. (accessed: 14 May 2023).
- [14] CSA, “CSA Cloud Controls Matrix (CCM).” <https://cloudsecurityalliance.org/research/cloud-controls-matrix/>, (accessed: 14 May 2023).
- [15] “5G Security Controls Matrix–ENISA.” <https://www.enisa.europa.eu/publications/5g-security-controls-matrix/>.
- [16] “free5GC.” <https://www.free5gc.org/>. (accessed: 14 May 2023).
- [17] K8s, “Kubernetes.” <https://kubernetes.io/>. (accessed: 14 May 2023).
- [18] “Towards5GS-helm.” <https://github.com/Orange-OpenSource/towards5gs-helm>, Dec. 2022. (accessed: 16 May 2023).
- [19] M. N. Hoq, J. W. Yao, S. Majumdar, L. Suárez, L. Wang, A. Boukhtouta, M. Pourzandi, and M. Debbabi, “Evaluating the security posture of 5g networks by combining state auditing and event monitoring,” in *European Symposium on Research in Computer Security*, Springer, 2023.
- [20] Ericsson, “5G Core (5GC) network: Get to the core of 5G.” <https://www.ericsson.com/en/core-network/5g-core>, Dec. 2022. (Dec 2024).
- [21] “Taxonomy of attacker capabilities.” <https://cloud.google.com/blog/products/identity-security/identifying-and-protecting-against-the-largest-dos-attacks>, May 2023. (accessed: 16 May 2023).
- [22] I. Nadir, Z. Ahmad, H. Mahmood, G. A. Shah, F. Shahzad, M. Umair, H. Khan, and U. Gulzar, “a1-an auditing framework for vulnerability analysis of IoT system,” in *2019 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, pp. 39–47, IEEE, 2019.
- [23] Mitre, “MITRE FiGHT.” <https://fight.mitre.org/>. (Dec 2024).
- [24] Mitre, “MITRE ATT&CK.” <https://attack.mitre.org/>, Dec. 2022.
- [25] N. Tamura, “Sugar: a SAT-based constraint solver.” <https://cpsat.gitlab.io/sugar/>, Dec. 2022. (accessed: 14 May 2023).
- [26] “ISO/IEC 27002:2022 Information security, cybersecurity and privacy protection—Information security controls.” <https://www.iso.org/standards/75652.html>, Dec. 2022. (accessed: 14 May 2023).
- [27] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [28] K. P. Murphy, *Dynamic bayesian networks: representation, inference and learning*. University of California, Berkeley, 2002.
- [29] pgmpy, “pgmpy 0.1.19 documentation.” <https://pgmpy.org/>, Dec. 2022.
- [30] graphviz, “Graphviz.” <https://graphviz.org/>, May 2023.
- [31] “NetworkX.” <https://networkx.org/>, Jan. 2023. (accessed: 16 May 2023).
- [32] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” 2019.
- [33] “bert-base-NER.” <https://huggingface.co/dslim/bert-base-NER>, 2023.
- [34] “PyDotPlus Homepage.” <https://pydotplus.readthedocs.io/>, May 2023.
- [35] “Open Source Cloud Computing Infrastructure - OpenStack.” <https://www.openstack.org/>, May 2023. (accessed: 16 May 2023).
- [36] 3GPPspace, “Inside TS 23.501: AMF Load Balancing,” Jan. 2021.
- [37] X. Ou, S. Govindavajhala, A. W. Appel, et al., “MulVAL: A logic-based network security analyzer,” in *USENIX security symposium*, vol. 8, pp. 113–128, Baltimore, MD, 2005.

- [38] S. Nie, Y. Zhang, T. Wan, H. Duan, and S. Li, "Measuring the deployment of 5G security enhancement," in *Proceedings of the 15th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2022.
- [39] M. Shafayat Oshman, *Assessing Security in the Multi-Stakeholder Premise of 5G: A Survey and an Adapted Security Metrics Approach*. PhD thesis, Carleton University, 2022.
- [40] S. Yang, D. Yin, X. Song, X. Dong, G. Manogaran, G. Mastorakis, C. X. Mavromoustakis, and J. M. Batalla, "Security situation assessment for massive MIMO systems for 5G communications," *Future Generation Computer Systems*, vol. 98, pp. 25–34, 2019.
- [41] M. Lei, L. Zhao, M. Pourzandi, and F. F. Moghaddam, "A hybrid decision-making approach to security metrics aggregation in cloud environments," in *2022 IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, pp. 178–185, IEEE, 2022.
- [42] X. Yang, "A Hidden Markov Model-Based Network Security Posture Prediction Model," *Applied Mathematics and Nonlinear Sciences*.
- [43] Y. Li, F. Wu, et al., "Improved Population Intelligence Algorithm and BP Neural Network for Network Security Posture Prediction," *International Journal of Distributed Sensor Networks*, vol. 2023, 2023.
- [44] D. Zhao, P. Shen, and S. Zeng, "ALSNAPE: Attention-based long and short-period network security situation prediction," *Ad Hoc Networks*, vol. 150, p. 103279, 2023.
- [45] A. Yang, B. Xie, Y. Liu, L. Wang, and J. Li, "A Network Security Situation Prediction for Consumer Data in the Internet of Things Using Variational Mode Decomposition (VMD) and fused CNN-BiLSTM-Attention," *IEEE Transactions on Consumer Electronics*, 2023.
- [46] T. Jiang and W. Liu, "Security Situation Assessment and Prediction of Power System Key Nodes," in *2022 First International Conference on Cyber-Energy Systems and Intelligent Energy (ICCSIE)*, IEEE, 2023.
- [47] J. Ma, Q. Xu, M. Zou, and J. You, "A Security Posture Prediction Method for Power Networks Based on Extreme Learning Machine," in *2023 5th International Conference on Electronic Engineering and Informatics (EEI)*, pp. 680–683, IEEE, 2023.
- [48] S. Bartoletti, G. Bernini, I. Palamà, M. De Angelis, L. M. Monteforte, T. E. Kennouche, K. Tsagkaris, G. Bianchi, and N. B. Melazzi, "Uncertainty quantification of 5G positioning as a location data analytics function," in *2022 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, IEEE, 2022.
- [49] M. Pendleton, R. Garcia-Lebron, J.-H. Cho, and S. Xu, "A survey on systems security metrics," *ACM Computing Surveys (CSUR)*, 2016.
- [50] S. Majumdar, G. S. Chawla, A. Alimohammadifar, T. Madi, Y. Jarraya, M. Pourzandi, L. Wang, and M. Debbabi, "ProSAS: Proactive security auditing system for clouds," *IEEE Transactions on Dependable and Secure Computing*, 2021.
- [51] K. Huang, X. Zhang, Y. Mu, F. Rezaeibaghia, X. Wang, J. Li, Q. Xia, and J. Qin, "EVA: Efficient versatile auditing scheme for IoT-based datamarket in jointcloud," *IEEE Internet of Things Journal*, vol. 7, no. 2, pp. 882–892, 2019.
- [52] G. Apruzzese, R. Vladimirov, A. Tastemirova, and P. Laskov, "Wild Networks: Exposure of 5G Network Infrastructures to Adversarial Examples," *IEEE Transactions on Network and Service Management*, 2022.
- [53] F. Boeira, M. Asplund, and M. Barcellos, "Provable Non-Frameability for 5G Lawful Interception," in *ACM WiSec*, 2023.
- [54] C. J. Mitchell, "The impact of quantum computing on real-world security: A 5G case study," *Computers & Security*, vol. 93, 2020.
- [55] M. Orsós, M. Kecskés, E. Kail, and A. Bánáti, "Log collection and SIEM for 5G SOC," in *2022 IEEE 20th Jubilee World Symposium on Applied Machine Intelligence and Informatics (SAMI)*, IEEE, 2022.
- [56] IBM, "IBM Security QRadar XDR.." <https://www.ibm.com/qradar>, Dec. 2022. (accessed: 14 May 2023).

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